DIAGNOSYSTEM, SURVEILLANCE AND MONETARY EXPENDITURE FOR EMAKAELA, TUPE AND EXPENDITURE
RECOMMENDATIONS FOR HPV-VASTAS VACCINATION
ENS Treatment Guide, version 4

Approved by the boards of the Estonian Society of Gynecologists, the Estonian Colposcopy Society, the Estonian Society of Family Physicians, the Estonian Midwives Society, the Estonian Society of Dermatologists, the Estonian Society of Pathologists and the Estonian Society of Laboratory Medicine

Compiled by: Inga Aavik, Lee Padrik, Terje Raud, Katrin Täär, Piret Veerus

The aim of the guide is to ensure the same evidence-based principles for diagnosing, monitoring and treating precancerous cervical changes in Estonia. The guide is aimed at gynaecologists, oncogynecologists, skin and sexually transmitted diseases doctors, general practitioners, midwives and residents of the above-listed specialties.

The guide provides an overview of the epidemiology of cervical, vaginal and vulva cancers, organised and opportunistic screening in Estonia. The guide includes recommendations for gynaecologists to monitor the positive outcome of organised screening, to monitor cervical changes in women aged 25-29 and to deal with situations that occur in a number of other cases. The guide discusses issues related to monitoring, treatment and post-treatment control of cervical, vaginal and vulva dysplasias and summarises the HPV vaccination situation in Estonia and further recommended directions.

The guide is recommended.

This guide replaces version 3 of the "Diagnosis, monitoring and treatment of precancerous cervical changes" approved by the Board of the Estonian Society of Gynaecologists.

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A. Abbreviations, evidence-based definition

1.1. Abbreviations

AIS (ld *adenocarcinoma in situ)*– early form of cylindrical-derived cancer

ASC-H (atypical *squamous cells, not excluded HSIL)*- Atypical squamous cells of indeterminate meaning for which high-grade intraepithelial lesions (HSIL) cannot be excluded

Atypical *squamous cells of undetermined significance* (ASCUS)

AGUS (atypical *glandular cells of undetermined significance)*

Atypical *glandular cells* (AGC)

AGC-NOS (atypical *glandular cells, not possibly reactive)*

AGC-FN (atypical *glandular cells favor neoplaasia)*– glandular cells of indeterminate meaning, probably neoplastic

CCI (clinical-colposcopic *index)*

CIN (cervical *intraepithelial neoplasia)*- cervical intraepithelial neoplasia divided into 3 grades

- CIN1 - Mild Grade Dysplasia
- CIN2 - Moderate dysplasia
- CIN3 - Severe Grade Dysplasia
- CIN1+ - Mild grade dyplasia and all more severe dysplastic changes
- CIN2+ - Moderate dysplasia and all more severe dysplastic changes
- CIN3+ - Severe grade dysplasia and all more severe dysplastic changes

CIS (ld *carcinoma in situ)*– early form of cancer

CKC (cold *knife conization)*

DES (Diethylstilbestrol) – Diethylstilbestrol

ECC (Endocervical *Curettage)*- Cervical Canal Abrasion

IFCPC (International *Federation for Cervical Pathology and Colposcopy)*– International Federation of Cervical Pathology and Colposcopy

Human *immunodeficiency virus* (HIV)

HPV (human *papillomavirus)*

hrHPV (high-risk *human papillomavirus)*- high-risk human papillomavirus

HSIL (high-grade *squamous intraepithelial lesion)*- high-grade squamous intraepithelial lesion

ISO (International *Organization for Standardization)*

LBC (liquid-based *cytology)*– liquid-based gynaecocytology

LEEP - *loop electrosurgical excision procedure*, synonymous with LLETZ in the USA

LLETZ (large *loop excision of the transformation zone)*

LSIL (low-grade *squamous intraepithelial lesion)*- low-grade squamous intraepithelial lesion

NAT study– test based on nucleic acid sequencing

NILM (negative *for intraepithelial lesion or malignancy)*- normal Pap response: no intraepithelial lesions or malignancies in the assay

Papanicolaou's *test*– conventional gynaecocytological test

SCJ (squamocolumnar *junction)*– a line between the flat and cylindrical epithelium in the cervix

SIL (squamous *intraepithelial lesion)*- squamous intraepithelial lesion

SLNB (Sentinel *node biopsy)*- Watchman lymph node biopsy

TBS (the *Bethesda system),* a system for describing cytological diagnoses in the cervix and vagina

'TZ' (transformation *zone)*

TTKS – Health Services Organisation Act

VAIN (vaginal *intraepithelial neoplasia)*- vaginal intraepithelial neoplasia or dysplasia

VAIN1 - Mild vaginal dysplasia

VAIN2 - Mean Vaginal Dysplasia

VAIN3 - Severe vaginal dysplasia

VIN (Vulva *intraepithelial neoplasia)*- Vulva intraepithelial neoplasia or dysplasia

VIN1-Vulva mild dysplasia

VIN2 - Mean Vulva Dysplasia

VIN3 - Vulva severe dysplasia

VSR - Cancer Screening Register

1.2. Definition of evidence-based action recommendations

A (High Evidence Quality Level) recommendation for action is based on the results of at least one correctly conducted randomised controlled trial or meta-analysis involving multiple trials.

The operational recommendation for B (Measurable Evidence Quality Level) is based on: (a) the results of a properly conducted, controlled, non-randomized study; (b) the results of a prospective or retrospective cohort study; (c) the results of a multi-centre, case-control study.

C (low level of evidentiary quality) is based on the results of a number of well-conducted independent monitoring studies, case descriptions and the translation of the results of the studies listed in the previous sections into conditions close to those examined.

D (Very low level of evidentiary quality) recommendation for action is based on expert judgement, clinical experience and good clinical practice, there are no studies listed in the previous sections.

The strength and weakness of the recommendations are not graduated in the guide, but this is reflected in the wording used. For example, the wording 'is indicative' indicates a strong pro-recommendation and the wording 'is allowed' indicates a weak pro-recommendation. For example, the wording 'not correct; not indicative' has been used in the case of a strong counter-recommendation and 'should not' has been used in the case of a weak counter-recommendation.

B. Diagnosis, monitoring and treatment of precancerous cervical conditions

1. Background

1.1. Epidemiology of cervical cancer in Estonia

Cervical cancer is the second most common gynecological malignancy in Estonia. Every year in Estonia, about 150 women are diagnosed with cervical cancer and about 70 women die from it (1).
Although national organised cervical cancer screening started in 2006 in Estonia, the incidence rate of cervical cancer in Estonia was the third highest in Europe in 2008-2012 (2). The age-adjusted incidence rate of cervical cancer in Estonia has risen by 0.8% every year since 1980 (Figure 1). The incidence has increased in all age groups, except for those over 70, and in all birth cohorts, except for women born in the 1940s. The proportion of advanced or stage IV cases of cervical cancer has increased from 13% between 2005 and 2009 to 18% between 2010 and 2014 (Figure 2). At the same time, the proportion of cancer cases detected in Stage I decreased from 40% to 35% (3).
The death rate from age-adjusted cervical cancer was the ninth highest in Europe in 2013 (4). Cervical cancer morbidity data show that the national screening programme has not produced the desired results (3, 5-7). Experience from other countries shows that with well-organised screening, both morbidity and mortality fall by 80% (8.9).

1.2. Cervical cancer screening in Estonia and elsewhere in Europe

In Estonia, organised cervical cancer screening was started in 2006 and in the same year only 31% of those invited participated in the survey. Cervical cancer screening in Estonia is organised by the Estonian Health Insurance Fund and the National Institute for Health Development, which are under the administration of the Ministry of Social Affairs. Pap as a first round test is still used to detect cervical cancer and precancerous changes in the cervix (10). Women aged 30 to 55 years without complaints are invited to cervical cancer screening at five-year intervals for findings from previous gynaecocytological examinations without pathology (NILM) for the entire duration of the screening target population. The cancer screening register managed by the National Institute for Health Development brings together the target group and the data to be invited to the screening. In the case of pathology detected during screening, further monitoring and treatment is carried out in accordance with the guidelines for the monitoring and treatment of precancerous changes in the cervix, vagina and vulva (11).

In January 2017, the Netherlands was the first country in Europe to switch to HPV-based cervical cancer screening among women aged 30-60, with the additional possibility of home testing (12). The percentage of positive findings increased from 5% to 9% and 3% of screening participants were directed to colposcopy instead of 1%. The detection rate of CIN2+ increased from 11 women to 14 per 1,000 screened. In 2019, Finland, Sweden, Italy and Turkey had switched to partial HPV-based cervical cancer screening (13). Estonia plans to

switch to HPV-based screening in 2021.

The Cervical Cancer Screening Registry was established in Estonia in 2015. The VSR publishes data on participation in screening, but not on the results of studies and additional studies, and on the treatment performed. According to the VSR, participation in cervical cancer screening has been between 46 and 51% among the target population in different years. Research has shown that an HPV test taken by a woman as a home test is as reliable as an HPV test taken by a healthcare professional (14) and that home testing can increase screening coverage (15).

1.3. Vaginal and vulva cancer epidemiology in Estonia

The incidence of pubic and vaginal cancer has been stable in Estonia, including among both younger and older women. Mortality of these tumors has also remained unchanged. The five-year relative survival rate for vulvar and vaginal cancer has improved and reached 61%. In Estonia, there is no more frequent trend in the incidence of younger women characteristic of Western Europe, which may be related to a lower prevalence of HPV in the population (16).

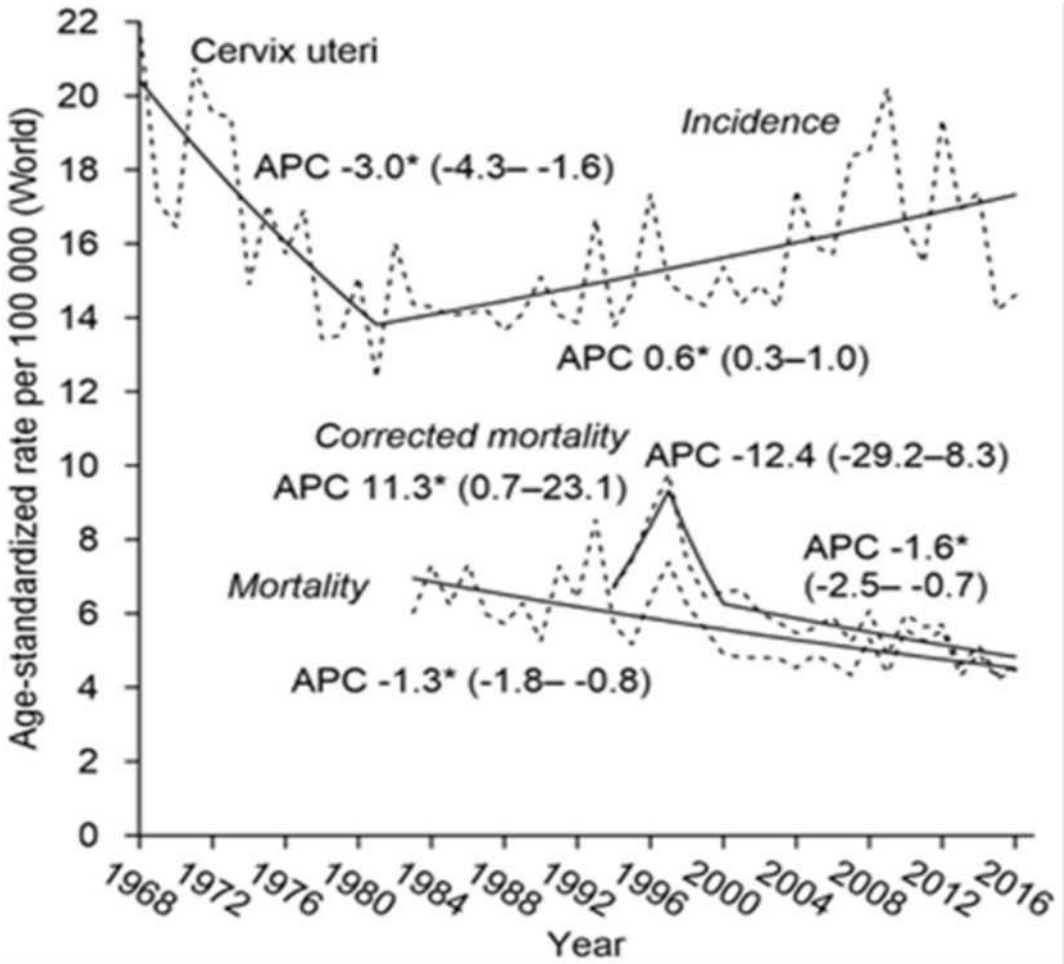
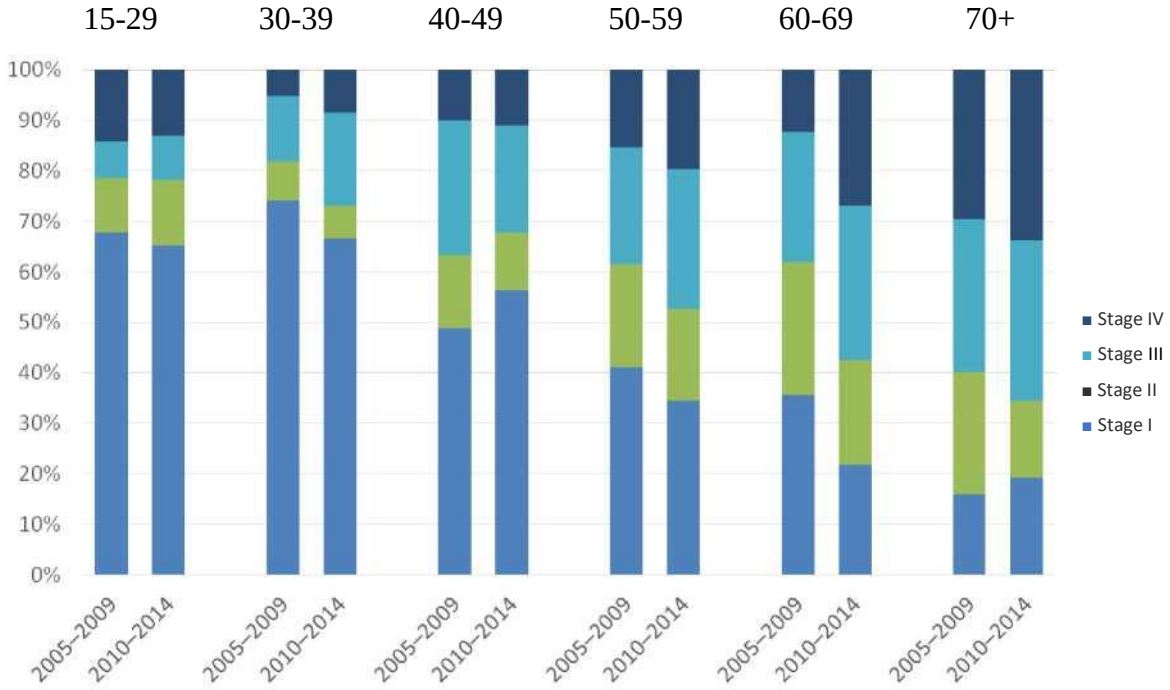


Figure 1: Standardised rate of cervical cancer morbidity and mortality per 100,000 women in Estonia between 1968 and 2016 (all age groups).

Figure 2: Prevalence of cervical cancer at first diagnosis by age group, 2005-2009 and 2010-2014



1.4. Course of HPV infection and association with cervical cancer

Genital HPV is transmitted through sexual and genital skin-to-skin contact. Perinatal, transplacental (17) and sexual transmission are also rarely possible. More than two-thirds of sexually active women are infected with a type of HPV during their lifetime (18). The incidence of HPV infection is highest in young people within a few years of the onset of sexual activity. In Estonia, the average age of young people aged 15-24 at the time of first sexual intercourse is 16.1 years (19) and the overall prevalence of HPV is highest in the 21-25 age group (20).
Approximately 90% of HPV infections are transient and heal on their own. A condition for the development of cervical cancer is a persistent infection caused by a high-risk HPV genotype. The global prevalence of HPV in cervical cancer is 99.7% (21). HPV type 16 is responsible for approximately 55-60% and the next most common HPV type 18 for 10-15% of cervical cancer cases (21-23). The next 12 high-risk HPV types in descending order for cervical cancer are 45, 31, 33, 52, 58, 35, 59, 56, 51, 39, 68 and 73 (24). The 14 high-risk HPV types listed account for 94% of all cervical cancers in the world.
Women with persistent infections caused by HPV16, HPV18, HPV31, HPV33 and especially HPV16 are significantly at risk of developing precancerous conditions. In Europe, the proportion of HPV16 in women with HSIL is 52%, and this is the most common type of HPV. HPV18, with a share of 6%, ranks fourth (25). The probability of developing a CIN3 or more severe change (CIN3+) with 2 years of HPV16 persistents is 26.7% (26) over the next 12 years. Untreated CIN3 has a 30% chance of progressing to cancer within 30 years. HPV is the cause of all common and most rare histological types of cervical cancer. HPV types 16, 18 and 45 account for 94% of all adenocarcinoma cases worldwide (21).

2. Methods of cervical examination and diagnosis

2.1. Gynecocytological examination

During the gynaecocytological examination, a preparation of a cell scraper taken from the vaginal wand, cervix and cervical canal with a spatula or brush and painted with Papanicolaou paint mixture for microscopic examination (Pap preparation) is examined. In Estonia, the Bethesda system and two methodologies are used for the assessment of gynaecocytological tests: conventional gynaecocytology (PAP) and liquid-based gynaecocytology (LBC).

2.1.1. Technique for carrying out conventional gynecocytology

The material is taken from ecto- and endotservics. One of the three methods (A) must be used (27).

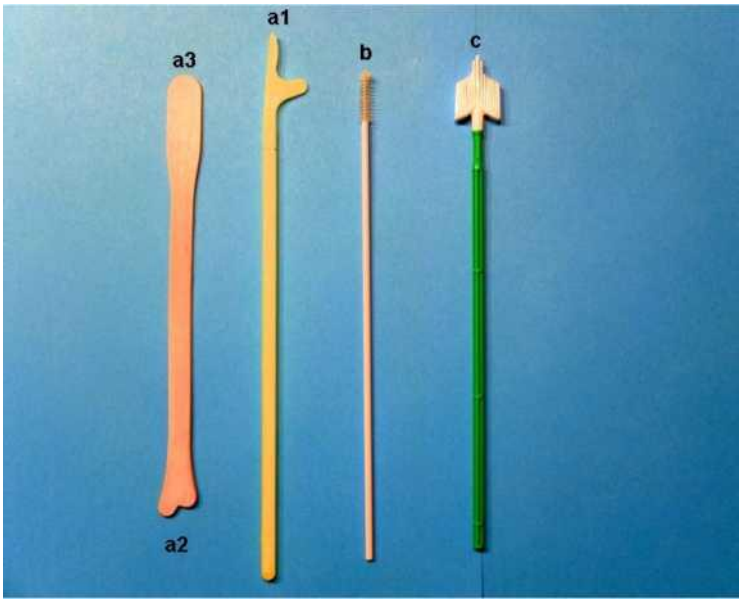


Figure 3: Applicators: (a) spatula: (a1) Aylesbury spatel with extended tip; (a2) Ayre spatel; (a3) round spatel, (b) endothelial brush, (c) cervical brush.

- Extended-peak Aylesbury Spa alone (Figure 3a1).
  The material is collected simultaneously from the endo- and ecto-server and applied longitudinally from both sides of the spatula to the glass.
- Combination of either Ayre spatula (Figure 3a2) or round spatula (Figure 3a3) ectotserviksi and a brush (Figure 3b) for endotservix.
  This is the best method if the transformation zone is located high in the neck canal, for example, in menopause, if there is a previous cervical surgical procedure, or if there is a pronounced ectropion. By maintaining constant contact with the neck and grasping the transformation zone, the spatula is pulled over the ecto-serve with more than a 360-degree circular motion. The brush is inserted 2/3 into the neck canal and moved about 90-180 degrees from the ring. The material is rolled from the brush in the opposite direction to the outer third of the glass in a smooth motion and pulled perpendicularly from the spatula to the middle third of the glass.
  The endocrine brush should never be used as the only tool.
- A cervical brush on its own (bromo; figure 3c).
  The best method for pregnant women and easily bleeding cervix.
  The brush is moved five times in a circular 360-degree clockwise direction, with longer bristles gripping the neck canal and shorter portios. The material is applied longitudinally from both sides of the brush to the glass.

Immediately fix the material by placing 95% ethyl alcohol in a glass container for at least 10 minutes, drip the solution onto the glass or use an aerosol fixator. In the latter case, the glass must be kept at a distance of at least 20 cm from the nozzle in order to prevent mechanical breakage and wiping of the cells from the slide (28).

2.1.2. Liquid-based gynaecocytology technique

The analysis is carried out in the same way as a conventional study. Rinse the collected material in the transport solution. In the laboratory, mucus, blood and other extraneous material are separated and the cells are applied in a single layer to the slide by centrifugation or filtration techniques. They are painted according to Papanicolaou and evaluated similarly to conventional research. It is possible to determine HPV analysis from the same study material, etc.

There is no significant difference in the sensitivity and specificity of liquid-based cytology and

conventional cytology to HSIL, but in liquid-based cytology the number of inadequate analyses is significantly lower (A) (29).

2.2.3. Quality and characteristics of gynaecocytological examination

All gynaecocytological examinations used in cervical surveillance in Estonia must be carried out in accordance with the requirements for pathology laboratories, which are described in the Code of Practice of Cervical Cancer Screening of the Estonian Health Insurance Fund (35). The requirements are as follows:

Requirements for the laboratory

- The pathology laboratory has been accredited to the ISO 15189 standard for liquid-based gynaecocytology (LBC) and conventional gynaecocytology (PAP) methodologies.
- Has the ability to perform gynaecocytological, histological and immunohistochemistry studies.
- Using the LBC methodology, HPV NAT analysis must also be available in the same facility.
- Effective intra-laboratory quality control in terms of both cytotechnics and diagnostics, i.e. rescrimination.
- Operational non-laboratory quality control in the LBC methodology – Participates in some internationally recognised quality control systems.
- The laboratory information system shall allow the flow of the responses to the digital lock and the flow of the slides operational extraction from the archives. The storage of data must be carried out in accordance with the applicable regulations (TTKS, Archives Act).
- The most recent Bethesda classification is used to formalise survey responses in cytology.
- A sufficient number of personnel must be ensured in the laboratory, taking into account the load requirements imposed on both the screening laboratory/bioanalyst and the cytologist/pathologist by the Estonian Society of Pathologists.
- Capabilities of laboratory/bioanalyst mastering cytotechnics are ensured, gynaecocytology the ability of the screening laboratory/bioanalyst and the ability of a cytologist/pathologist competent in gynaecology.

Personnel requirements

- Laboratory/Bioanalyst – has both LBC and CT competencies EAA methodology.
- Screener – holds a qualification as a bioanalyst or laboratory assistant, or in the field of natural sciences have a university degree, adequate training and work experience in cervical cancer screening, have at least the annual workload required to maintain their qualifications.
- Cytologist/pathologist – qualified as a pathologist or cytologist, trained and Professional experience in cervical cancer screening, with at least the annual workload required to maintain the qualification.

Factors affecting the quality of gynaecocytological examination include:

- Menstruation, bleeding in the genital tract
- Vaginal and cervical inflammation
- Sexual intercourse within 24 hours before analysis

Factors affecting the interpretation of gynaecocytological examination are:

- Severe vaginal atrophy (menopause)
- Pregnancy and the postpartum period and lactation period, the cause of reactive change
- Previous manipulation or chemical irritation: bimanual vaginal observation, lubricant, ultrasonic gel, disinfectant cream, spermicide, vaginal medicine, vaginal rinse, spermicide or acetic acid test on colposcopy (less than 24 hours ago), previous gynecological examination (less than 3 weeks ago), cervical surgery (less than 3 months ago)
- Radiation therapy

According to the Bethesda system, the test is assessed as adequate or inadequate. Preparations where more than 75% of the material is concealed by blood, inflammatory material or drying defects that are broken or where identification of the patient is not possible shall be considered inadequate and the reason for the inadequacy shall be indicated in the response.

Only the presence of cylindrical epithelium cells in the preparation makes it insufficient, and the clinician must then decide whether to take a new preprate. Regardless of the outcome of the HPV test (unknown, negative or positive), it is recommended to repeat the gynaecocytological test after 2-4 months (D) if the result of the gynaecocytological test is not adequate. An indicator of transformation zone involvement is the presence of endothelial cells or metaplastic cells.

Preparations without involvement of the transformation zone (no cylindrical epithelial or metaplastic cells) shall be considered adequate and the analysis need not be repeated. Long-term studies have not shown an increased risk of severe grade dysplasia or cancer in women whose gynaecocytology has not included the transformation zone (30). The increased risk of dysplasia and cervical cancer in women treated with CIN2+ is not associated with the absence of a tranformation zone in a gynecocytological study (31).

The sensitivity and specificity of the gynaecocytological examination are not known. Sensitivity to CIN2 and more severe changes at the low cytological threshold is 70-80%, which is relatively high. At the high cytological threshold, sensitivity decreases and specificity increases.

The high specificity of the gynaecocytological study allows to assess the immediate risk of CIN3+, but unlike the HPV study, the low sensitivity and low negative predictive value of the gynaecocytological study do not allow to predict the long-term risk of CIN3+ (32).

Combination data from all European and North American studies (over 60,000 women) comparing the pre-existing and specificity of gynaecocytology and HPV as compared to CIN2+ gave gynaecocytology a sensitivity of 53% and a specificity of 96% (33).

2.2. HPV study

The understanding of the link between the high risk of HPV infection and the risk of developing cervical cancer has led to the development of molecular HPV assays.

In Estonia, HPV NAT has been used as the primary test for cervical cancer screening since 2021.
In cervical cancer screening, only the determination of high-risk HPV is relevant.
Low-risk HPV is not associated with an increased risk of developing cervical cancer and does not play a clinical role in cervical cancer screening or in the evaluation of women with abnormal cytology.
Under the recommendations for the determination of HPV analysis, the text will always refer to the determination of hrHPV.

- ***.2.1.Quality, properties and use of the HPV study***

All HPV diagnostics used in clinical practice must meet the Meijer and co-authors validation criteria (34) and the requirements of the Estonian Health Insurance Fund's Code of Practice on Cervical Cancer Screening for HPV NAT diagnostics (35). Requirements:

- The HPV diagnostic test system used for cervical cancer screening must be: validated according to the requirements of the *European Guidelines for Quality Assurance in Cervical Cancer Screening* referring to the Meijer criteria.
- Use a set of hrHPV (high-risk genotypes) reagents that can differentiate HPV genotypes 16 and 18 and detect at least the following HPV genotypes: 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, and preferably additional genotypes 66, 68.

Compared to the gynaecocytological study, the HPV study has a higher sensitivity to CIN2+ and CIN3+, but a lower specificity and is more reproducible. Randomized studies show significantly lower cumulative incidence of CIN3+ and even cervical cancer in women 30 years and older who were initially HPV DNA-negative compared to Pap or LBC-negative women. (A)

There is sufficient evidence that the hrHPV study can be used in the following cases (A):

- HPV study in primary screening alone or in combination with a gynaecocytological study;
- ASCUS monitoring;
- post-treatment monitoring of cervical dysplasias

The HPV test should not be used:

- screening of women under 30 years of age as a stand-alone study or in combination with a gynaecocytological study;
- screening for other cancers and genital warts caused by HPV in men and women

2.3. Colposcopy

Colposcopy is a study of the cervix, vagina and vulva using a colposcopy. The colposcopic is a binocular microscope that allows 6-40 fold magnification of the cervix, vagina and vulva epithelium and blood vessels. The aim of the colposcopic study is to locate and localize suspicious lesions, assess their extent and severity, take a target biopsy and thereby help in choosing the right monitoring or treatment tactics.

Randomized biopsies are not indicated for cases of HPV16/18 negativity, gynaecocytology ≤ LSIL, adequate colposcopy and normal colposcopic findings, as the risk for this group is between 1 and 7% for CIN2+ and less than 1% for CIN3+ (36).

2.3.1. Conducting a colposcopic examination

Colposcopies are performed by a gynaecologist whose number of colposcopies for both low and high grade changes is >50 per year (37).

It is important to inform the patient about the need for the study and to describe the course of the study. This will ensure better cooperation between the colposcopyist and the patient. The best time to do a colposcopic examination is on the 10th-14th day of the menstrual cycle. 24 hours before the test is not recommended:

- use of vaginal drugs, lubricants and spermicides;
- do vaginal rinsing;
- have vaginal sexual intercourse;
- take tests from the cervix

The cervix must be fully visible. If the side walls of the vagina interfere fully with the visualization of the cervix, it is necessary to use lateral vaginal retractors. Before the start of the colposcopic examination, the cervix may be cleaned, if necessary, with either a saline solution or a 3% soda solution.

To visualize the abnormal finding of the cervix, 3-5% acetic acid solution is used, the action time is 1 minute.

The use of Lugol solution in colposcopic examination is recommended.

2.3.2. Indications for colposcopic examination

- Gynecocytological study ASC-H, HSIL, AGC-NOS, AGC-FN, AIS or repeated ASCUS and LSIL
- HPV pos and gynaecocytology ≥ LSIL
- HPV Pos and Gynecocytology ASCUS Repeatedly
- monitoring of cervical changes found
- Visible lesions on the cervix, vulval, vaginal mucosa
- Unexplained genital haemorrhages, contact haemorrhages

2.3.3. Contraindications to colposcopic examination

Absolute contraindications are absent and relative contraindications are

- acute inflammation of the genital tract
- menstruation

2.3.4. Complications of colposcopic examination

- occur extremely rarely
- sometimes bleeding is possible after a target biopsy (more frequent during pregnancy or after extensive excision)

2.3.5. Adequacy of colposcopic examination

- the colposcopic examination is adequate (diagnostic) if the colposcopy shows: the original cervical squamous epithelium and the cylindrical epithelium and the boundary between them (SCJ) to the full extent; the cervical transformation zone to the full extent; the existing abnormalities to the full extent; if the boundary between the cylindrical epithelium and the squamous epithelium is visible by an endocervical mirror;
- A colposcopic examination is considered to be inadequate (non-diagnostic) if any of the four previous criteria are not met.

2.3.6. Terminology established by the IFCPC for use in colposcopic examination

Overall assessment		Adequacy of the colposcopy 1. Adequate 2. Inadequate Type of TZ – 1, 2, 3.

Normal colposcopy finding		Original (original) flat epithelium, mature flat epithelium, atrophy Cylinder epithelium, ectopia Metaplastic flat epithelium, ovula Nabothi, open glands
Pathological colposcopy finding	General principles	Lesion localization - TZ inside or outside, localize by quadrant and clock dial position Lecture size - how many quadrants will cover
	Grade I – Low grade change (weak change)	Thin acetowhite epithelium, irregular (geographic map-like) edge Regular fine mosaic and puncture
	Grade II – High grade change (strong change)	Matte white acetowhite epithelium. Rapid appearance of acetowhite area Irregular coarse mosaic and puncture Sharp edges, hearth inside

Non-specific changes	Leukoplaakia (keratosis, hyperkeratosis), erosion
Suspicion of invasion	Atypical blood vessels. In addition, it easily breaks brittle blood vessels, uneven lesion surface, exophytic surface, necrosis, ulceration, visible tumour
Miscellaneous changes	Congenital TZ, condylomas, polyp, stenosis, post-treatment change, congenital anomalies, endometriosis, inflammation

2.3.7 Assessment of the colposcopic study

There are various colposcopic finding evaluation systems in use around the world. The most well-known of these are the Reid colposcopy index (Table 1), the modified Reid index (Table 2) and the Swede index (Table 3).According to the EFC recommendation, it is also recommended to use the Swede colposcopy index in Estonia.

The edges, color and vascular system of the cervical transformation zone or lesion in the cervix and, if necessary, the discoloration with iodine shall be evaluated. In addition to the four classical parameters (coloration with acetic acid, coloration with iodine, the outline of the hearth and the drawing of blood vessels), the Swede colposcopy index also assesses the size of the lesion. Findings are scored on a scale of 0, 1 and 2. The aggregated score correlates well with the severity of the change in the cervix.

We recommend using the standard colposcopy protocol in Estonia, which has been developed by combining the terminology established by the IFCPC and the assessment of the Swede colposcopy index.

2.3.9. Standard protocol for colposcopy

1. Adequacy of the colposcopy
   1.1. Adequate
   1.2. Inadequate
2. Type of Transformation Zone
   2.1. Type 1
   2.2. Type 2
   2.3. Type 3
3. Nature of the colposcopic find
   3.1. The Normal Find
   3.1.1. Mature flat epithelium
   3.1.2. Atrophy
   3.1.3. Ectopic
   3.1.4. Metaplastic flat epithelium
   3.1.5. Ovula Nabothi
   3.1.6. Open glands

   3.2. Pathological findings
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3. Recommendations for cervical examination

3.1. Age limits

The benefits and harms of performing an HPV or gynaecocytological examination for women of different ages are different. Therefore, the recommendations depend on the age of the woman (38).

3.1.2. Start of study

Cervical cancer is rare among adolescent and young women, so according to the IARC (International *Agency for Cancer Research),* it is not reasonable to start organised cervical cancer screening before the age of 25. The incidence of cervical cancer in the 20-24 age group is 1.4/100 000 women per year (39), and studies have shown that the risk of cervical cancer is not lower in women screened in this age group than in unscreened women. Screening of teenage and young women leads to the diagnosis and treatment of these dysplasias, with a high probability of spontaneous regression. In addition, surgical treatment of

cervical dysplasia increases the risk to reproductive health and is therefore potentially harmful (33, 40). Rapidly progressive cervical cancer in teenagers and young women cannot be prevented by screening (41). Prevention programmes for cervical cancer in teenagers should focus on HPV vaccination that is safe, high-effective (A)(42) and cost-effective before sexual activity.

3.1.3. End of study

Monitoring of women over the age of 65 may be discontinued if previous studies have been negative and the woman has not had a history of CIN2 or a more severe change in the past 25 years. Previous negative monitoring means 3 negative gynaecocytological tests or 2 negative HPV tests in the last 10 years, the last of which has been given within the last 5 years. For women above the observational age who have not been tested before or whose results are unknown, it is recommended to confirm with 2 negative HPV tests (interval of 5 years) or 3 negative gynaecocytological tests if the woman does not wish to perform an HPV test. Once the monitoring is complete, it is not right to restart it for any reason, even if the woman has a new sexual partner. A new high-risk HPV infection in women over 65 years of age should spontaneously heal in most cases, and only a small number of women become persistent (43). Based on the natural course of HPV infection, it is unlikely that CIN3 that develops after the age of 65 will have enough time to progress to cancer over a woman's lifetime. In the older population, the risks of overtreatment outweigh the benefits of screening. Women with a history of CIN2, CIN3 or AIS should continue to be tested for at least 25 years, even if testing exceeds the age limit of 65, as the risk of developing cervical cancer is about 5-10 times higher than in the general population (44).

3.2. Interval of studies

3.2.2. Recommendations for women aged 25-29

For women between 25 and 29 years of age, it is recommended that only a gynaecocytological examination be carried out every 3 years.

Since there is no strong evidence base, it is not right to screen women of any age annually with any method. Annual screening leads to a large number of unnecessary examinations and treatments, but increases the number of preventable cancers very little (34.45). Randomized studies do not provide a sufficiently high evidence base to increase the screening interval beyond 3 years in any age group in case of a prior negative gynaecocytological examination. Due to the high prevalence of HPV among women under 30 years of age (46), as confirmed by a survey conducted in Estonia (20), it is not appropriate to screen women under 30 years of age (25-29 years of age) for HPV.

3.2.3. Recommendations for women aged 30-65

For women between 30 and 65 years of age, an HPV study is recommended every 5 years.

A clinically validated HPV study in women 30 years of age and older has been shown to be more effective than a gynaecocytological study in primary screening, and the screening interval for HPV-negative women can be safely extended to at least 5 years (A). Randomized studies show significantly lower cumulative incidence of CIN3+ and even cervical cancer in women 30 years and older who were initially HPV DNA-negative compared to LBC or Pap-negative women (A). Screening with HPV alone provides the same level of protection as twin screening (HPV and gynaecocytology), as the risk of CIN3+ and cancer in women is only slightly lower if both tests are negative (HPV and gynaecocytology) compared to women with negative HPV and positive gynaecocytology (47). Compared to the detection of squamous cell cancer, cytology has been more ineffective in detecting adenocarcinoma (A)(48). Testing with HPV NAT also increases the likelihood of detecting adenocarcinoma and its pre-existing changes. If an HPV study is used in opportunistic screening, the 5-year interval should be strictly monitored, as more frequent testing increases the number of unnecessary colposcopy and treatment procedures (45, 47).

3.3. Control of women after hysterectomy

If the hysterectomy is performed with the removal of the cervix and there is no history of CIN2 or more severe change, no further monitoring is necessary.

After supraservic hysterectomy, the woman is subject to routine monitoring for cervical cancer.

If the hysterectomy is performed due to CIN, the monitoring tactics are the same as after cervical consolation:
Twin study 6 and 24 months after treatment (49, 50).

- If the tests are negative, repeat the twin test after 3a (B). If the twin study is negative, continue routine (HPV-based) monitoring for at least 25 years (D)
- If the twin study is HPV positive at any time or ≥ LSIL is found in gynaecocytology, a colposcopy should be performed. In case of ASCUS, perform repeated twinning after 12 months

Vaginal cancer is a very rare gynecological tumour, therefore screening for vaginal cancer is not justified.

4. Monitoring and tactics in case of cervical changes

4.1. Monitoring of the results of organised screening by a gynaecologist

Figure 5:

Since 2021, an HPV/NAT study (35) has been organised in Estonia for the primary screening of cervical cancer in women ≥ 30 years of age.
If the primary test result of organised screening is negative for HPV, routine monitoring in screening continues at 5 year intervals.
Opportunistic use of HPV between organised screening intervals is not recommended (51).
In the event of a positive result in any HPV screening test, the LBC test shall be performed immediately as part of the screening.

Re-examination of female HPV-positive patients ≥ 30 years is monitored by gynaecologists outside of screening (35).
If the result of the primary screening test is:

- HPV positive and LBC ≥LSIL, directs the primary investigator to female colposcopic examination (35)
- HPV positive and LBC NILM or ASCUS, the primary investigator recommends repeating the HPV test after 12 months (52) at the gynaecologist's appointment

If in the first re-examination

- HPV negative, routine screening continues
- HPV positive and LBC ≥ ASCUS, directed by gynaecologist to colposcopic examination of woman
- HPV positive and LBC = NILM, the HPV test is repeated at a gynaecologist's appointment after 12 months.

If in the second re-examination after 24 months

- HPV negative, routine screening continues
- HPV positive and LBC ≥ LSIL, directed by gynaecologist to colposcopic examination of woman
- HPV positive and LBC = ASCUS, repeat HPV study after 12 months at gynaecologist

appointment

Colposcopy is certainly performed in cases where HPV positivity first occurs and the LBC response is ≥LSIL or where the combination of HPV pos and ASCUS has persisted for 12 months

4.2. Monitoring of the cervix based on the results of the gynaecocytological examination

4.2.2. Monitoring of women ≥ 30 years of age based on gynaecocytological findings

4.2.2.1 Gynecocytological test response ASCUS - Figure 6.

In women who have ASCUS in the gynaecocytology study, it is recommended to perform an HPV study or it is allowed to repeat gynaecocytology after 12 months:

- In case of a negative HPV response, routine monitoring in screening continues
- In case of a positive HPV test response, it is recommended to repeat the HPV test after 12 months and to implement further monitoring of HPV positive women according to the scheme described above.
- In case of repetition of gynaecocytological examination, refer the woman to colposcopy if the cytological result worsens during 12-month follow-up periods, i.e. is ≥LSIL, or if ASCUS occurs in the second repetition, i.e. the change has persisted for 2 years

4.2.2.2 Gynecocytological test response LSIL - Figure 7.

It is recommended to perform an HPV test:

- In case of a positive response to LSIL and HPV or if the HPV status is unknown, colposcopic testing is recommended. Further monitoring according to further histological findings (see CIN monitoring)
- In case of a negative LSIL and HPV response, routine monitoring in screening continues

4.2.2.3 Gynecocytological test response ASC-H - Figure 9.

Irrespective of HPV status, colposcopic testing is recommended for ASC-H. HPV testing prior to colposcopic examination is not recommended (B).

4.2.2.4 Response of the gynaecocytological examination HSIL - figure 9.

For HSIL, colposcopic examination (B) is recommended. Further therapeutic tactics according to histological diagnosis of biopsy.

Tracking tactics by repeating an HPV or gynaecocytological examination are not allowed.

4.2.3. Monitoring of women aged 25-29

Outside organised screening, women between 25 and 29 years of age may be monitored at 3 year intervals and gynaecocytological examination is recommended as the primary test.

4.2.3.1 Gynecocytological test response ASCUS - Figure 8.

The preferred tactic is to repeat the gynaecocytological examination after 12 months. Immediate colposcopy and HPV testing are not recommended.

In case of repetition of gynaecocytological examination:

- If ASCUS or LSIL is the result of a gynaecocytological examination after 12 months, cervical monitoring should be continued with a gynaecocytological examination at 12 months intervals.
- Refer the woman to colposcopy if gynaecocytological outcome worsens during 12 months of follow-up, i.e. ≥ASC-H or LSIL persists for more than 24 months

- If after 12 months the response of the gynaecocytological examination is NILM, then routine monitoring will continue (repeat the gynaecocytological examination after 3 years)
- If a woman has reached the 30-year limit during the follow-up period, cervical monitoring will continue according to the monitoring scheme for women ≥30 years of age

4.2.3.2 Gynecocytological test response LSIL - Figure 8.

It is recommended to monitor on the basis of a gynaecocytological examination. The HPV study is not recommended

- Refer the woman to colposcopy if gynaecocytological outcome worsens during 12 months of follow-up, i.e. ≥ASC-H or LSIL persists for more than 24 months
- If after 12 months the response of the gynaecocytological examination is NILM, then routine monitoring will continue (repeat the gynaecocytological examination after 3 years)
- If a woman has reached the 30-year limit during the follow-up period, cervical monitoring will continue according to the monitoring scheme for women ≥30 years of age

4.2.3.3 Gynecocytological test response ASC-H or HSIL - Figure 9.

It is recommended to conduct a colposcopic examination (B). Immediate excision with a sling is not allowed (B). Further therapeutic tactics according to histological diagnosis of biopsy.

4.2.4. Monitoring of cylindrical epithelium changes in all age groups

4.2.4.1 Gynecocytological test response AGC figure 10.

Tracking tactics for AGC are the same for all age groups (B).

- If AGC-NOS was present in the gynaecocytological study, an HPV study and a colposcopic study with cervical channel material are recommended for further monitoring
- If AGC-FN was present in the gynaecocytological study, it is recommended that colposcopy be supplemented with a histological examination of the uterine cavity

Histological analysis of the uterine cavity (with colposcopy and histological analysis of the cervical canal) is recommended for all women over 35 years of age (B).
A histological examination of the uterine cavity (with colposcopy and cervical canal analysis) is recommended for women under 35 years of age if they are at risk of uterine neoplasia. The risk group includes women with unexplained vaginal bleeding and signs of chronic anovulation.

4.2.4.2 Gynecocytological test response AIS - Figure 11.

If AIS is diagnosed, it is recommended to consult an oncogynecologist for treatment tactics

- In women who are still planning to give birth, cervical constipation is the first recommended method of treatment

The cervical consolation is sufficient if the cut lines of the histological preparation are clean (endocervical). If CIN or AIS changes occur at the cutting edges of the histological preparation, it is recommended to repeat the conisation.

- In women who no longer plan to give birth, the primary method of treatment is hysterectomy.

4.2.5. Suspicion of invasive carcinoma

If possible, perform a colposcopy with an excisional biopsy or immediately refer the patient to an oncogynecologist.

4.2.6. Monitoring of benign glandular changes

In premenopausal asymptomatic women who had benign endometrial cells, endometrial stroma cells or histiocytes in a cytological test, further studies are not recommended (B).
In postmenopausal asymptomatic women, uterine cavity control (B) is also recommended for benign endometrial cells.
For women with hysterectomy and benign glandular cells in the gynaecocytological response, further studies are not recommended (B).

4.2.7. Monitoring of women with immunosuppressive conditions - Figure 12.

Immunosuppressive conditions are HIV disease, post-organ transplantation or post-allogeneic stem cell transplantation, systemic lupus erythematosus, ulcerative colitis and rheumatic diseases requiring immunosuppressive therapy.
Regardless of the age of the woman, it is recommended that cervical monitoring be started within a year of the start of sexual activity and continued until the end of life.

<30-year-old women gynaecocytological examination once a year for 3 years, further gynaecocytological examination at 3 year intervals
Because HIV-positive women have a relatively high prevalence of hrHPV, monitoring of <30-year-old women by double-study is not recommended (53).

30% of women who are perinatally infected with HIV have ASCUS (54) as a result of the primary gynaecocytological examination.

- If the response of the primary gynaecocytological examination is ASCUS, the examination should be repeated after 12 months
- If the retest response is ≥ ASCUS, refer the woman to colposcopy
- If the primary gynaecocytological test response is ≥LSIL, refer the woman to colposcopy

In women ≥ 30 years of age, twin study (HPV and gynaecocytological study) at 3 year intervals throughout life

HIV-positive women have a high risk of CIN3 + change in low-grade cytological outcomes (54).

- If the HPV test is negative and the gynaecocytological test response is NILM, the binary test interval is 3 years
- If the HPV test is negative and the gynaecocytological test response ≥ LSIL, refer to colposcopy
- If the HPV test is positive and the gynaecocytological test response ≥NILM, refer to colposcopy

4.3. Monitoring of the cervix in other cases

4.3.2. Clinical indication for the study

Visible lesions in the cervix, genital bleeds, contact bleeds

- At the age of <30, choose a gynaecocytological examination for the primary examination
- Dual study (HPV and gynaecocytology) **is allowed at ≥30 years of** age

4.3.3. Women ≥ 30 years of age, not participating in organised screening

Select HPV as the primary study

4.3.4. >65 year old woman, not screened

It is recommended that a negative result be confirmed in women who have exceeded the observation limit by age and who have not been tested before or whose results are unknown.

- 2 negative HPV tests (interval of 5 years) or
- 3 negative gynecocytologic examinations (interval 3 years) if the woman does not wish to perform an HPV examination

4.3.5. Monitoring transgender men

In transgender men with preserved cervix, cervical examinations should be carried out in the same way as for women. Since transgender people with a man's personal identification code are not invited to cervical cancer screening, it is important not to forget about the necessary studies for them, which must be carried out throughout life in the same way as women. As an alternative to HPV testing by a healthcare professional, they may be offered HPV home testing.
The risk of cancer in the cervix, vagina and vulva of transgender women depends on the method by which vaginoplasty has been performed. The need for research is decided individually.

5. **Monitoring and treatment of cervical dysplasias**

5.1. Monitoring the results of colposcopy and planning treatment

In general, CIN1 patients do not require treatment as up to 80% of such lesions regress spontaneously within 2 years (55), <15% CIN1 progresses to CIN2/3 within 2 years (56), so most cases of CIN1 do not progress to severe grade dysplasia.

Bansal and co-authors (56) showed in their retrospective study that within 6 months CIN1 resolves in 49% of women, 35% persists, and 7% of women develop HSIL (CIN2/3). At 12 months, women who maintained CIN1 at month 6 had a further 50% reduction, 46% remained, and 4% of women had developed HSIL lesions. If colposcopy due to HPV positive, ASCUS/LSIL had no lesion or histological incidence of CIN1, the relative risk of developing CIN3+ within 5 years is only 2% (57).

If CIN1 lesion persists for more than two years, cervical condensation may be performed (58).

Since the risk of CIN2 and CIN3 progression is high, treatment for these changes is recommended. CIN2 is 22% more likely to progress to CIN3 and 5% more likely to progress to invasive cancer (59). CIN3 is up to 40% (60) more likely to progress to invasive cancer if left untreated. Spontaneous regression can occur in 32-47% of cases. (B).

In the case of CIN2, the desire of a woman to become pregnant plays an important role in the decision on monitoring and treatment tactics, as cervical formation is associated with some risks to reproductive health.

CIN3 should generally be treated in all pregnant women.

According to the Code of Practice for Testing for HIV Infection and Referral to Treatment of HIV-positive Persons, it is recommended to test all women with pre-cancerous conditions of the cervix, cervical cancer and anal cancer for the presence of HIV virus.

5.1.2. CIN1

- .1.1.1. HPV Positive or Unknown, in Gynecocytology ASCUS, LSIL. Adequate colposcopy, histology ≤ CIN1 - Figure 13
    - Repeat the HPV test after 12 months (for women under 30 years of age, gynaecocytological

examination only) (B).

- If the HPV test (a gynaecocytological test in women under 30 years of age is NILM) is negative,
  Repeat the HPV test (genocytological examination less than 30 years) after 3 years.
- If the HPV test is positive, perform an additional gynaecocytological test and if the response is:
  - LSIL, direct the woman to a colposcopy. If the answer to the gynaecocytological examination is ASCUS, repeat the HPV examination (for women under 30 years of age, gynaecocytological examination only) after 12 months.

If CIN1 persists during the 2 year follow-up period, follow-up tactics may be continued or cervical consolation may be performed.

- .1.1.2. HPV positive or unknown, in gynecocytology ASC-H or HSIL. Adequate colposcopy, histology ≤ CIN1 - Figure 14

It is advisable to choose one of three options:

- If the HPV is positive/unknown and there is a finding of ASC-H or HSIL in gynecocytology, but there is no lesion in CIN 2+ on colposcopy, it is recommended to review the results of cytology, colposcopy and histology (B).
- Diagnostic Consistency
- Monitoring tactics for 2 years by repeating an HPV study (under 30 years only gynaecocytological study) after 12 and 24 months.

  - If the HPV (gynecocytological) test is negative on both occasions, the next HPV (gynecocytological) test may be repeated 3 years after the last test.
  - If any HPV study taken during the follow-up period is positive or ≥ LSIL is detected in the gynaecocytological study, a repeat colposcopy should be performed.

If the response to the HPV test remains positive during the 2 year follow-up period and a finding of HSIL/ASC-H in the gynaecocytological study is observed, cervical retention is recommended. If CIN1 persists over the 2 year follow-up period, taking into account the low (2%) risk of CIN3+ over the next 5 years, you can continue with monitoring tactics or perform cervical concentration.

ASC-H: Tracking tactics are preferred for up to 2 years

HSIL: can carry out cervical concentration, but can also apply monitoring tactics for up to 2 years.

When choosing tactics, proceed from the patient's preference (acceptance of repeated visits during the follow-up period), desire to become pregnant.

5.1.3. Histological study CIN2, CIN2/3 or CIN3 - Figure 15

If CIN3 is found in the colposcopy biopsy or cervical abrasion material, cervical consolation should be performed (except for pregnant – see corresponding section).

In case of a finding of CIN2 in a non-pregnant patient, it is also recommended to perform cervical consolation, but can also apply monitoring tactics in certain cases.

The goal of the monitoring tactics is to reduce the risks to reproductive health in women planning future pregnancies (e.g. premature birth due to possible cervical insufficiency due to congestion).
In young women under 30 years of age, the proportion of spontaneous regression of CIN2 is high. Based on a meta-analysis and systematic review carried out in 2018, CIN2 resolves in up to 60% of cases in women under 30 years of age, 23% persists and 11% progresses, so monitoring tactics could be applied in unborn women in case of CIN2 finding (61).

Monitoring tactics can be applied in case of an adequate colposcopic finding for up to 2 years, first repeating a 6-monthly HPV study (under 30 years) and a colposcopy.

- If HPV is positive in 2 consecutive assays or if ASCUS or LSIL occurs in a woman under 30 years of age in a gynecological study, but the finding is less severe than CIN2 in a histological

study, repeat the HPV study (a gynecological study under 30 years of age) after 12 months. If the response is negative, continue testing 1x per year for 3 years, and if the response is negative, continue monitoring with a standard cervical cancer screening interval of at least 25 years.

- If, after 6 and 12 months from the primary study, HPV is positive/gynecocytology ASC-H/HSIL/AGC and a CIN2 or milder change persists on colposcopy, follow-up should be continued at 6-month intervals for up to 2 years from the primary CIN2 diagnosis.

- If CIN2 persists after 2 years or if CIN3 has formed in any intervening period, it is recommended to perform cervical consolation.

5.1.4. Pregnant women and cervical dysplasia

The tactics of monitoring cervical changes during pregnancy are similar to those of non-pregnant women. Pregnant women do not have a higher risk of precancerous cervical changes or cervical cancer compared to non-pregnant women.

Colposcopy with cervical biopsy (portiolt) is allowed during pregnancy, but cervical abrasion or endometrial biopsy should not be performed.

In case of suspicion of slight grade change (in Gynecocytology ASCUS/LSIL), postponement of colposcopy to the postpartum period is allowed.

CIN1 detected during pregnancy does not require intervention as the risk of CIN2/3 and cervical cancer is low and the proportion of CIN1 regression after childbirth is high (62).

HSIL (CIN2/3) detected during pregnancy aims to exclude invasive cervical cancer. It is recommended that pregnant women with CIN2/3 undergo a gynaecocytological examination and colposcopy at 12-24 week intervals (B) until the end of the second trimester (30 weeks of pregnancy). A repeated biopsy is only required if there is a suspicion of invasive cancer in gynaecocytology or if the colposcopic finding worsens. Re-examinations for CIN 2/3 during pregnancy may also be postponed to the postpartum period – either by HPV or gynaecocytological examination and colposcopy no earlier than 6 weeks after childbirth, depending on age (B).

Diagnostic cervical condensation during pregnancy is allowed only in case of suspicion of invasive cancer.

5.1.5. If for any reason a woman under 25 years of age (21-24 years of age) has been studied

Since women under the age of 25 have a high rate of spontaneous regression of changes in HPV infection, CIN1 and even CIN2, and a low incidence of cancer, monitoring tactics are applied to these changes. According to the United States, the incidence of cervical cancer in this age group is 1.4 per 100,000 women per year (50).

5.1.5.1 Gynecocytology study response in < women 25 years of age with LSIL, histology ≤ CIN1:

- Gynaecocytological examination at 12 months (HPV study not allowed (B));
- If there is an ASCUS/LSIL response in the gynaecocytological study, repeat the study after 12 months;
- if after 12 months in the gynaecocytological response ASC-H/HSIL, refer to colposcopy;
- if after 12 months of gynaecocytological examination in response to NILM, continue with normal monitoring (B)

5.1.5.2 Gynecocytological response in < women 25 years of age ASC-H, HSIL, histology ≤ CIN1:

In case of adequate colposcopy, follow-up tactics within 24 months are recommended. Carry out a

gynaecocytological examination at 12-month intervals.
- if found during the follow-up periodin the gynaecocytological study ASCUS/LSIL, repeat gynaecocytological examination after 12 months
- if ASC-H/HSIL is detected during the observation period, refer to colposcopy
- if 2 consecutive gynaecocytological examinations and colposcopy are negative during the follow-up period, continue with routine monitoring;
- if after 24 months HSIL persists, but there is no lesion referring to CIN2+ in the colposcopy, it is recommended to perform a diagnostic cervical concentration (B);
- if the colposcopy is inadequate, diagnostic cervical condensation (B) may be performed.

5.1.5.3 Gynecocytological test response < in women 25 years of age HSIL (CIN2, CIN2/3 or CIN3)

Tracking tactics are recommended for CIN2.

Adequate colposcopic findings and histological HSIL (without specifying either CIN2 or CIN3) may be subject to monitoring tactics.

With histologically confirmed CIN3, cervical condensation can be performed.

In case of adequate colposcopy, cervical monitoring is allowed for 24 months (52), carrying out both gynaecological examination and colposcopy at 6-monthly intervals for 12 months (D). If during the follow-up period 2 consecutive gynaecocytological test responses are negative (NILM) and the colposcopy does not visualize the lesion, perform the gynaecocytological test and colposcopy after 12 months (D).
If CIN2/3 has persisted during the 24-month follow-up period, cervical consolation (B) is recommended.
In case of inadequate colposcopy and CIN3, cervical consolation (B) is recommended.

5.2. CIN Therapy Methods

For women with cervical dysplasia, it is important to choose tactics that will help prevent the possible progression of dysplasia to cervical cancer, while avoiding overtreatment in case of changes that are likely to regress themselves. The need for and method of treatment of precancerous conditions of the cervix depends on the results of HPV, cytological, colposcopic and histological (biopsy) examinations and on the patient (age, presence of pregnancy/desire to become pregnant, therapeutic propensity).

For CIN1, the chance of progression to cervical cancer is minimal, while for severe grade dysplasia, the risk of cancer is high. CIN3 has a 30% risk of progressing to cervical cancer within 20 years (53). In general, women with cervical dysplasia should be treated with CIN2+. The goal of treatment is to remove the cervical transformation zone and with it the lesion (dysplastic epithelium).
Ablative and excitation methods are used as treatment methods. Excision is the preferred method of treatment, as it makes it possible to obtain material for histological examination and thereby establish a correct diagnosis and assess the purity of the cut lines. At the same time, the therapeutic efficacy of both ablative and excitation methods is similarly close to 90% (63).

a) Ablative (destructive) methods:
- cryotherapy
- laser vaporisation
- diathermocoagulation

Conditions for Ablative (Destructive) Treatment:
- adequate colposcopic examination (total TZ visible);
- Fully visualized lesion;
- histological diagnosis excluding cylindrical epithelium involvement and invasion;
- the cytological and histological diagnosis are the same;
- superficial lesion, diameter < 2 cm, not involving more than a quarter of the cervical surface;

• Adequate equipment is required.

b) Methods of Excitation
• Cervical Consulation:
- – LEEP/LLETZ with electrical link (most commonly used) – treatment effectiveness 94-98% (53);
- laser;
- with scalpel - CKC - recommended for invasive disease, cervical lesion/cylinder epithelial dysplasia.

The purpose of cervical continuum is to remove dysplastic epithelium. The depth of the conisate in the case of ecto-servic lesion should be 7-10 mm. If the depth of the cervical conisate exceeds 10 mm, the risk of premature birth increases. At a depth of more than 12 mm, the risk of premature birth is 3 x higher. At a depth of more than 20 mm, the risk of premature birth even rises to 18% (64).

Elsewhere in the world, the so-called *'see and treat'* (53) methodology is also used in certain cases, where a cervical configuration is performed in local anesthesia during the same visit as the colposcopy. The method is intended for use in cases where colposcopy clearly presents with CIN2+ cervical dysplasia or is suspected in a patient with a poor therapeutic propensity who is unlikely to be treated later. The advantage of the method is the possibility of immediate treatment of the precancerous condition of the cervix, so the number of visits is reduced, and patients who are otherwise ill will receive treatment in time. However, the application of the method increases the risk of overtreatment, as it has been shown that in nearly 20% of women who have undergone cervical consolation, no dysplasia was actually found in the cervical conisate. This method of treatment is not the first choice, but given the specific case of the disease, it can be used in women over 30 years of age (65).

- Cervical amputation

- Hysterectomy (if there is a concomitant gynecological pathology, in which the removal of the uterine body is also indicated, or if the lesion is located in the cervical canal)

Preferably, excision methods are recommended, since the result of this treatment is also a histological examination of the material, which makes it possible to assess whether the entire lesion has been removed (purity of the cut lines). If possible, the entire lesion should be removed in one piece. It is recommended to carry out cervical consolation in the first phase of the menstrual cycle.

5.3. Treatment of CIN1

Among the methods of treatment, both ablation and cervical continuum are suitable for an adequate colposcopic finding. If colposcopy is inadequate, or CIN2/3 is found in cervical abrasion, or the degree of dysplasia is not detectable, then cervical consolation is the recommended method of treatment. The treatment of hysterectomy CIN1 is not allowed (66).

5.4. Treatment of CIN2 and CIN3

The preferred method of treatment is cervical formation. For CIN2/3, hysterectomy is contraindicated as a first-line treatment (B)(67).

5.5. Post-treatment monitoring of CIN2/3

All women who have been treated for cervical dysplasia are at risk of recurrence of dysplasia and require regular postoperative monitoring. Women treated for cervical dysplasia have a 4-fold higher risk of cervical cancer (68) than those without a CIN. In 10% of treated women, CIN2+ recurs. After treatment, the risk of developing cervical cancer remains elevated for at least 20 years (69). After ineffective treatment, recurrent dysplasia appears most often within 2 years after treatment (70). Tracking tactics are influenced by the applied treatment method – it is not possible to assess the cleanliness of the cut lines after ablative treatment. In the case of incomplete excision/ablation, CIN2+ occurs later in up to 18%, in the case of complete excision in 3% of treated women (71). CINs are more likely to recur if the

cutting lines are involved, but this is not the only risk factor (72). Relapses of cervical dysplasia occur more frequently in older women (> 30 years), previously treated for severe degree dysplasia, and in women who smoke. Also in women with lesions in the cervical canal who remain infected with HPV despite treatment.

Both the HPV test and the twin test are equivalent to predict a CIN2+ relapse after cervical concentration.

The risk of CIN2+ is 0.8-1.4% (73), 0.68% in the double study and 2.4% (74) in the gynaecocytological study in women who tested negative for HPV after concentration.

Women who have a negative twin count after 6 months of cervical condensation have a CIN2+ risk of less than 0.5% (75) in the next 2 years.

According to a study conducted in the Netherlands, the probability of recurrence of CIN2+ after 5 years of treatment for cervical dysplasia in women with a negative twin count at 6, 12 and 24 months post-treatment is 1%, after 10 years it is 3.6%. The probability of developing CIN2+ at 6, 12 and 24 months is 2.9% at 5 years and 5.2% at 10 years in women who have been monitored by gynaecocytology alone (76).

5.5.1. Recommendations for CIN post-treatment monitoring

Twin study 6 and 24 months after treatment (50, 69).

If the results of the studies are negative, repeat the double study after 3 years (B). If the twin study is negative, continue routine monitoring (based on HPV) for at least 25 years (D).
If the response to the HPV test is positive in any of the twin studies or ≥ LSIL is found in the gynaecocytological study, a colposcopy should be performed. In the case of ASCUS, perform repeated twinning after 12 months - if ASCUS persists, repeat the colposcopy.
If the dysplastic epithelium reached the intersection line, repeated concentration (D) may be performed.
In case of a CIN2+ relapse, a concentration or hysterectomy (B) may be performed.

C. Diagnosis and treatment of precancerous conditions of the vagina and vulva

1. Vaginal precancerous conditions

1.1. Nature and causes of precancerous vaginal conditions

It is not known exactly, but the causes and risk factors are likely to be similar to those of cervical and vulva dysplasias. The most common cause is HPV infection, 100% of women with mild vaginal dysplasia are HPV positive, 90% with moderate and severe vaginal dysplasia. HPV16 occurs in 53-57% of severe cases of vaginal dysplasia and vaginal cancer (77). Vaginal dysplasia is significantly more common in women (50-90% of women with VAIN) who have previously been diagnosed with cervical or vulva dysplasia or malignant disease. After hysterectomy, women with a history of CIN2+ change have a 5% risk of vaginal dysplasia and dysplasia may develop over a long period (4-13 years) after surgery(78, 79).

The potential for VAIN to progress to vaginal cancer is unknown. No prospective studies have been conducted, some retrospective studies have shown that 2-8% of VAIN cases progress to invasive vaginal cancer.

Most lesions (90%) are located in the upper third of the vagina, and in 50% of cases are multifocal.

Vaginal dysplasia usually does not cause complaints, and the vast majority of vaginal dysplasias are diagnosed through a response to a changed gynecological examination. Postcoital bleeding or

increased discharge may occur rarely. If vaginal dysplasia is suspected, a vaginal colposcopic examination is indicated.

1.2. Classification of vaginal dysplasias

Since 2012, a new classification of vaginal intraepithelial dysplasias is recommended as part of the LAST project:

- low-grade VAIN (intraepithelial lesion of vaginal flat epithelium LSIL, formerly VAIN 1)
- high degree VAIN (vaginal flat epithelial intraepithelial lesion HSIL, formerly VAIN 2-3)

Classification used before 2012 and still in use at times:

VAIN1 - Mild vaginal dysplasia

VAIN2 - moderate vaginal dysplasia

VAIN3 - vaginal severe dysplasia

1.3. Indications for vaginal colposcopy and vaginal colposcopy technique

- There are changes in the gynecological examination, but the cervix is healthy, there is no cervix, or cervical colposcopy not correlated with gynaecocytological test response
- Palpable, visible vaginal lesion
- Immune-suppressed patients with confirmed cervical or vulva lesions
- Previously treated premalignant or malignant vaginal lesions
- Unclear vaginal discharge
- Unclear bleeding
- Unclear pain, dyspareunia
- In utero DES (*diethylstilbestrol*) effects in foetus
- Before the planned concentration, if there are differences in gynaecocytology, cervical biopsy histology and colposcopy

Vaginal colposcopy technique is not significantly different from cervical colposcopy. To better highlight the lesions, 3-5% acetic acid solution is used, but compared to cervical examination, the duration of action should be longer, 3-5 minutes. Not all vaginal lesions become visible after treatment with acetic acid, so 50-60% Lugol solution is recommended for vaginal colposcopy. A biopsy should be taken from visible lesions using biopsy tubes (cervical) or a small coagulation ring.

Most lesions can be found in the upper third of the vagina, more often on the back (vaginal vault) or front wall, mosaic drawing is rare (visible in CTZ and adenomatosis), characteristic lesions with a rough, spiked surface, so-called peaks.

Since 2011, vaginal clinical/colposcopic terminology has been used, see Table 4.

Table 4: Colposcopic find characteristic of vaginal dysplasias

Characteristic of low-grade vaginal dysplasia	Characteristic of high-grade vaginal dysplasia
• Multi-bottom • Condylomas (vulva → 30% also vaginal) • Exquisite diffuse scoring • Diffuse small acetowhites areas • Descriptive iodine-negative areas (turtle shield pattern) • Small iodine-negative spikes, peaks	• Single-bottom • Strongly aceto-white, with hints of pearl • No vascular drawing • Clear regular boundaries • Above the surface • Iodine-negative (mustard colour) • Epithelium hatched (in case of atrophy) • Atypical blood vessels – possible Sign of malignancy

1.4. Monitoring and treatment of vaginal dysplasias

Vaginal cancer is a very rare gynecological tumour, therefore screening for vaginal cancer is not justified. Tracking tactics with HPV and gynaecocytology are applied in exceptional cases (see 3.3. Control of women after hysterectomy).

Low-grade VAIN regresses spontaneously in most cases, high-grade VAIN is considered a precancerous condition. In 5 years, 78% of all vaginal dysplasias are regressed, VAIN persists in 13% of cases, and invasive cancer develops in 9% of cases (74).

- Low-level VAIN if applicable, implement observing tactics:twin study (HPV+gynecocytology) 1x per year for 2 years. If either HPV is positive or ≥ LSIL is detected in gynaecocytology in any of the twin studies, a colposcopy should be performed. For HPV neg and ASCUS, repeat the twin study after 12 months. If ASCUS persists, it should be directed to colposcopy.
  If the HPV test is negative during 2 years of follow-up and the gynaecocytological response is NILM, then further follow-up is the same as after hysterectomy (see 3.3. Control of women after hysterectomy).

- For high-grade VAIN, apply treatment. The choice of treatment method depends on the size, severity and age of the lesion. The preferred method is surgery. Other treatments - ablation (laser therapy, ultrasound) and drug therapy (preferably locally imiquimod 3 times a week for 8 weeks or 5-fluorouracil) are indicated if there is a multi-focal change and there is no suspicion of invasion, or if there is no suspicion of invasion and surgical excision leads to a significant anatomical change and dysfunction. Radiation therapy is applied in extreme cases, when other methods of treatment have failed, they are very large lesions, surgical treatment is contraindicated.

After treatment, long-term monitoring (at least 25 years) is required to exclude relapses (75).

VAIN post-treatment monitoring:

- Vaginal examination for assessment of vaginal mucosa every 6 months for 2 years after treatment and then once a year
- Once a year preferably twin study(HPV+gynecocytology).
- Colposcopy should be performed for any positive test (HPV positive or gynaecocytology ≥ ASCUS).

2. Precancerous conditions of Vulva

2.1. Epidemiology of Vulva's precancerous conditions

Vulva's precancerous states are intraepithelial lesions of vulva based on the flat epithelium - vulva SIL (formerly called vulva intraepithelial neoplasia - VIN) and non-lapepithelial vulva Paget disease and vulva melanoma in situ.

Screening methods for early detection of vulva precancerous conditions and vulva cancer are not known and routine screening is not performed.

Flat epithelial-derived dysplasia is significantly more common. The true incidence of vulva SIL is precisely unknown, as vulva SIL can often be asymptomatic and go unnoticed, but it is known that the incidence of vulva SIL has increased significantly in recent decades and is also more common in young women. Among women under the age of 35, the number of cases of vulva SIL has tripled. While before 1970, vulva SIL was mostly diagnosed between the ages of 50 and 60, today about half of patients diagnosed with vulva SIL are under the age of 40. Recent studies show that the incidence of vulva SIL increases up to the age of 40-49 and then gradually decreases, while the incidence of vulva cancer, on the contrary, increases up to the age of 60-69 (80, 81). In young women, vulva SIL is often multi-coloured and is caused by HPV. Approximately 80% of vulva dysplasia patients are HPV positive, 77.7% of vulva dysplasias are associated with HPV16 (82).

2.2. Classification of VIN

The classification of Vulva SIL has been changed several times over time. The most recent and widely used classification was recommended in 2015 by ISSVD (International *Society for Study of Vulvovaginal Disease).*

The classification of non-squamous vulva dysplasias has not changed.

2015 ISSVD vulva SIL classification:

2.2.1. *Lameracous changes*

- Low degree vulva intraepithelial lesion LSIL - vulva LSIL

Year	1986 ISSVD	2004 ISSVD	2012 LAST	2015 ISSVD
Terminology	VIN 1	genital warts (HPV effect)	LSIL	Vulva LSIL
	VIN 2 VIN 3	VIN common type: wart, basaloid, mixed type	HSIL	Vulva HSIL
	Differentiated VIN	Differentiated VIN	No not classified	Differentiated VIN

This includes changes in VIN1 according to previous terminology, genital warts, HPV infection, not considered precancerous

- High degree vulva intraepithelial lesion HSIL - vulva HSIL

This includes changes in HPV-associated VIN of the common type (typical and basaloid, mixed type), VIN2 and VIN3 according to previous terminology.

- Differentiated VIN (dVIN) - lesions caused by chronic skin diseases (sclerotic lichen, flat lichen, chronic simple lichen, etc.) not associated with HPV

2.2.2. *Non-squamous changes*

2.2.3. Paget's disease
2.2.4. Melanoma in situ

Vulva LSIL - benign skin changes due to HPV infection, which often heal spontaneously within 1-2 years. Not considered precancerous.

Vulva HSIL mostly occurs at a younger age (30-40 years), is associated with HPV infection, and smoking significantly increases the risk of developing it. Basaloid vulva HSIL regresses less frequently than warts and may progress more frequently to invasive cancer. Vulva HSIL is mostly asymptomatic, multi-coloured, lesions are often higher than the skin surface, rough and of changed color - white, brown, reddish.

Differentiated VIN develops in older people as a single-leaved process against the background of chronic skin diseases (lichen sclerosus, lichen simplex chronicus, lichen planus, etc.). 80% of Vulva squamous cell cancers are based on differentiated VIN. Lichen sclerosus patients have a 5% risk of developing vulva cancer (83). Differentiated VIN is often symptomatic, causing itching, tingling, pain. Often there are erosions, dense infiltrates, hyperpigmented areas.

Comparison of Vulva flat epithelial intraepithelial lesions (vulva SIL)

Histological form	**HPV**	**Not associated with HPV**
	Vulva HSIL	**Differentiated VIN**
Age	20-40 years	> 40 a
HPV	+++	-
Warts	+	-
Number of visits	Multi-bottom	Single-bottom
Immunosuppression, HIV	+	-
Link with chronic dermatosis	-	+
Cytological test changes	+	-
Molecular markers	p16 Ki67	p53
Smoking	+	-
Forecast	favorable	unfavourable

2.3. Indications and technique of Vulva colposcopy

- Surface lesions
- Mucous membrane, skin color changes: white, red, hyperpigmented areas
- Erosive, ulcerative areas
- Suspected genital warts that do not respond to conservative treatment
- Lichen sclerosus and squamous cell hyperplasia due to chronic skin diseases that do not respond to medication
- Suspicion of an invasive process for other reasons

Similar to colposcopy, 3-5% acetic acid is used, the duration of action is longer, 2 -5 min.

Vulva colposcopy makes it harder to assess lesions than cervical colposcopy. Thicker, hair-covered skin does not develop typical aceto-white areas with mosaic or dottedness. Such changes can be seen

only in thinner areas without hair (small pubic humps). All vulva lesions should be biopsied to clarify the diagnosis. In case of suspicion of melanoma, it is recommended not to take a biopsy, but to immediately carry out a broad excision within healthy tissues (1-2 cm), in case of a large-scale lesion, taking a biopsy is allowed. (A)

2.4. Treatment of Vulva dysplasias

- Vulva LSIL - often self-healing, treated with symptomatic foci or for cosmetic purposes. If necessary, treatment tactics appropriate to vulva genital warts are used.

- Vulva HSIL and DVIN are mostly treatable, but monitoring tactics may also be used in very exceptional cases, such as in patients with temporarily reduced immunity (pregnant women, women who have recently stopped corticosteroid therapy or smoking, etc.).

Different methods are used for treatment:

- **surgical excision within healthy tissues (0.5-1 cm), laser excision, laser ablation, vulvectomy.**

Surgical excision is the preferred treatment for differential VIN, Paget's disease and suspected melanoma.
Not enough research has been done to tell which surgical treatment method to prefer in different lesions. Based on literature data, different surgical treatments appear to have similar efficacy: relapse rate after vulvectomy 19%, partial vulvectomy 18%, local error 22% and laser ablation 23% (84).

- **medicamentous - imiquimod and orphan 5-FU (fluorouracil)**

Studies have shown complete improvement of 51%, partial improvement of 25% and relapses of 16% of those treated with imiquimod (85).

The risk of relapses after treatment with VIN is high throughout life.

In a study of 800 patients with vulva HSIL, 26% had relapses after treatment, of which 8% had invasive tumours and 25% had delayed relapses (44-196 months after initial treatment) (86).

Follow-up inspection (visual inspection of the volcano) is required 6 and 12 months after initial treatment and annually thereafter to rule out the occurrence of relapses.

- For VIN1, if necessary, apply treatment tactics appropriate to vulva genital warts
- VIN2 and VIN3 are mostly treatable, but monitoring tactics may also be used in very exceptional cases, such as in patients with temporarily reduced immunity (pregnant women, women who have recently stopped corticosteroid therapy or smoking, etc.).

Different methods are used for treatment:

- - surgical (preferred) cutting within healthy tissues (0,5-1 cm), laser excision, laser ablation, vulvectomy. Surgical excision is the preferred treatment for the suspicion of differentiated VIN, Paget's disease and melanoma;
- -Medical imiquimod and orphan 5-FU (fluorouracil) and interferon alfa.

The risk of relapses after treatment with VIN is high - 39% after local excision, 70% after laser ablation, 90% after 5-FU. Good results have been obtained in the treatment of VIN2 and VIN3 with imiquimod - up to 90% improvement (87).

Follow-up monitoring is required 6 and 12 months after initial treatment and annually thereafter for an

extended period of time to exclude relapses.

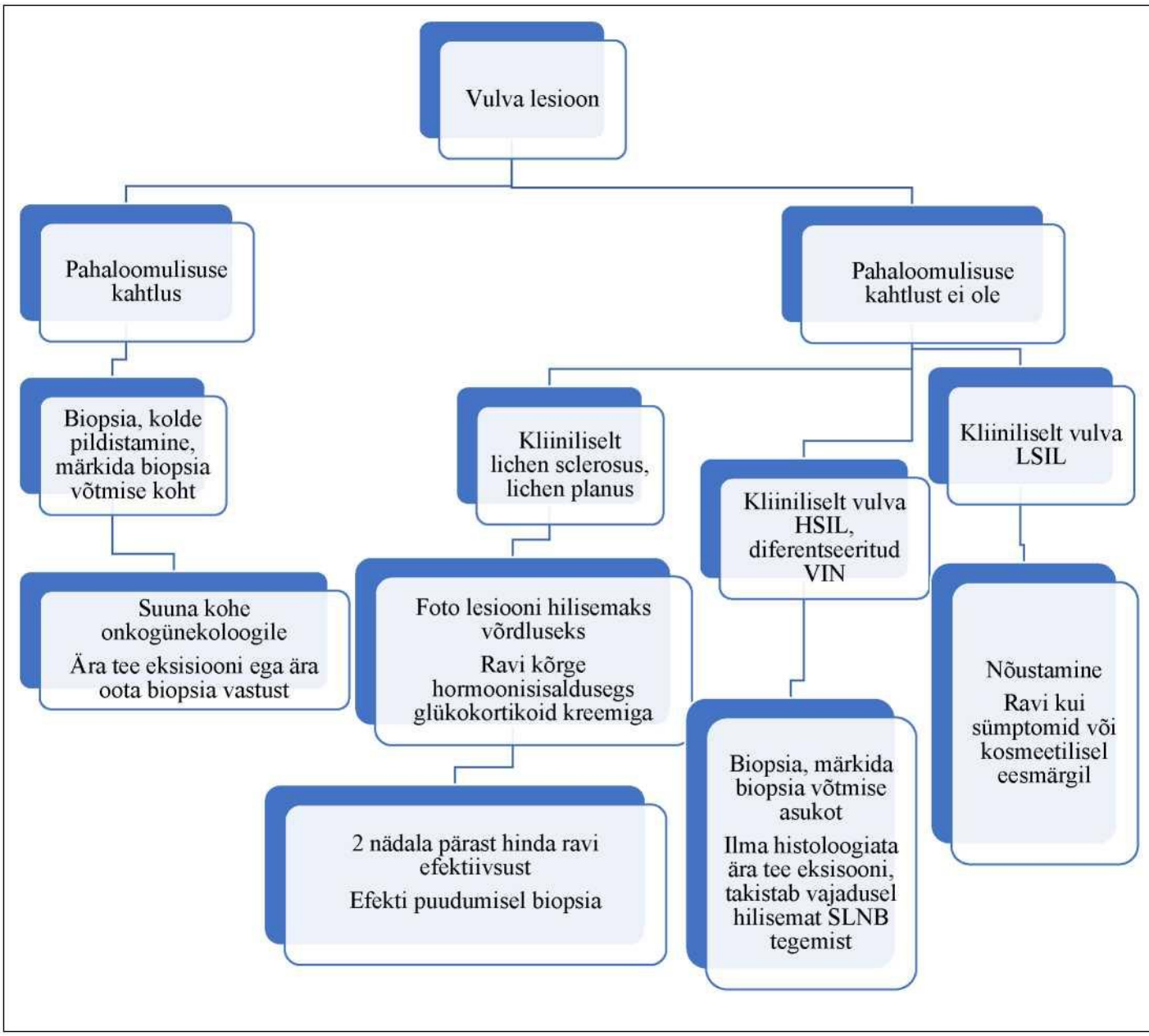

Figure 16: Vulva lesion treatment scheme

D. HPV vaccination

All HPV vaccines: bivalent, quadrivalent and ninevalent are safe, highly immunogenic and effective (88).

In women, HPV infection causes cervical, pubic, vaginal and anal cancers, precancerous lesions in these areas, and genital myeloma.

According to WHO recommendations issued in 2017, girls aged 9-14 years are the primary target group for cervical cancer prevention. Achieving a high level of coverage in this age group should be a priority for vaccination. High (>80%) coverage in vaccination in girls reduces the risk of HPV infection in boys (89).

Secondary target group: vaccination of women or men ≥15 years of age is recommended if it is

feasible and cost-effective for the country and does not deprive the primary target group of vaccination or an effective cervical cancer screening programme (89).

Vaccination of girls aged 9-14 years with 2 dose schemes is cost-effective (88).

Vaccination of girls against HPV is part of the national immunisation programme in Estonia. In 2018, vaccinations for girls aged 12-14 were started through a school programme. From 2020, only girls aged 12 years will continue to be vaccinated**.**
Of those born in 2004, 3,800 (62.6%) girls have been vaccinated with one vaccine dose and 60.9% with two doses. Of those born in 2006, 4,158 (63.4%) girls have been vaccinated with one vaccine dose and 54.1% (90) with two doses.

A meta-analysis of randomised controlled trials showed that vaccinating young women aged 16 years or older, regardless of their HPV status, reduced their risk of developing CIN2 or a more severe change due to HPV by 46% (91).

In order to increase the impact of the vaccination programme more rapidly, it is recommended to continue the catch-up group vaccination (ages 13-14) within the framework of the national programme and, if possible, to extend the group to the age of 18 years (89).

ENS recommends vaccinating all persons between the ages of 12 and 26 who have not been vaccinated through a national programme. Vaccination is allowed for women over the age of 26.

Post-cervical cervical condensation HPV vaccination significantly reduces the risk of new severe intraepithelial changes - RR 0.41 (95% UV 0.27-0.64) (92)(B). Recommend post-contamination vaccination.

Vaccination against HPV does not completely exclude the need for cervical cancer screening later in life, as existing vaccines do not provide protection against all high-risk HPV types and the protective effect of vaccination is lower in older age groups (89).

Coordination between the cervical cancer screening programme and the HPV vaccination programme and the creation of IT solutions between databases are important. At the time a woman reaches the screening target age group, the screening registry database should contain information on her HPV vaccination status (93).

In men, HPV infection causes head and neck cancers, anal cancer, penile cancer and precancerous lesions in these areas, as well as genital myeloma. In men, two-, four- and nine-valent vaccines against HPV are licensed for the prevention of precancerous lesions of the anogenital region and anal cancer. The total incidence of HPV-induced cancers of the oropharynx, oral cavity and larynx in men was 32,000 in 2018. It is the second most common HPV-induced cancer in the world after cervical cancer, with an increasing trend (94). Gender-neutral vaccination is unlikely to be cost-effective if the vaccination coverage of girls exceeds 50% (95).

ENS recommends the inclusion of HPV vaccination for 12-year-old boys in the national immunisation programme, based on the principle of individual benefit and gender equality.

E. List of literature and annexes

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2.Additions

2.1. Drawings on cervical monitoring

Figure 5: Monitoring of women with HPV positive ≥30 years

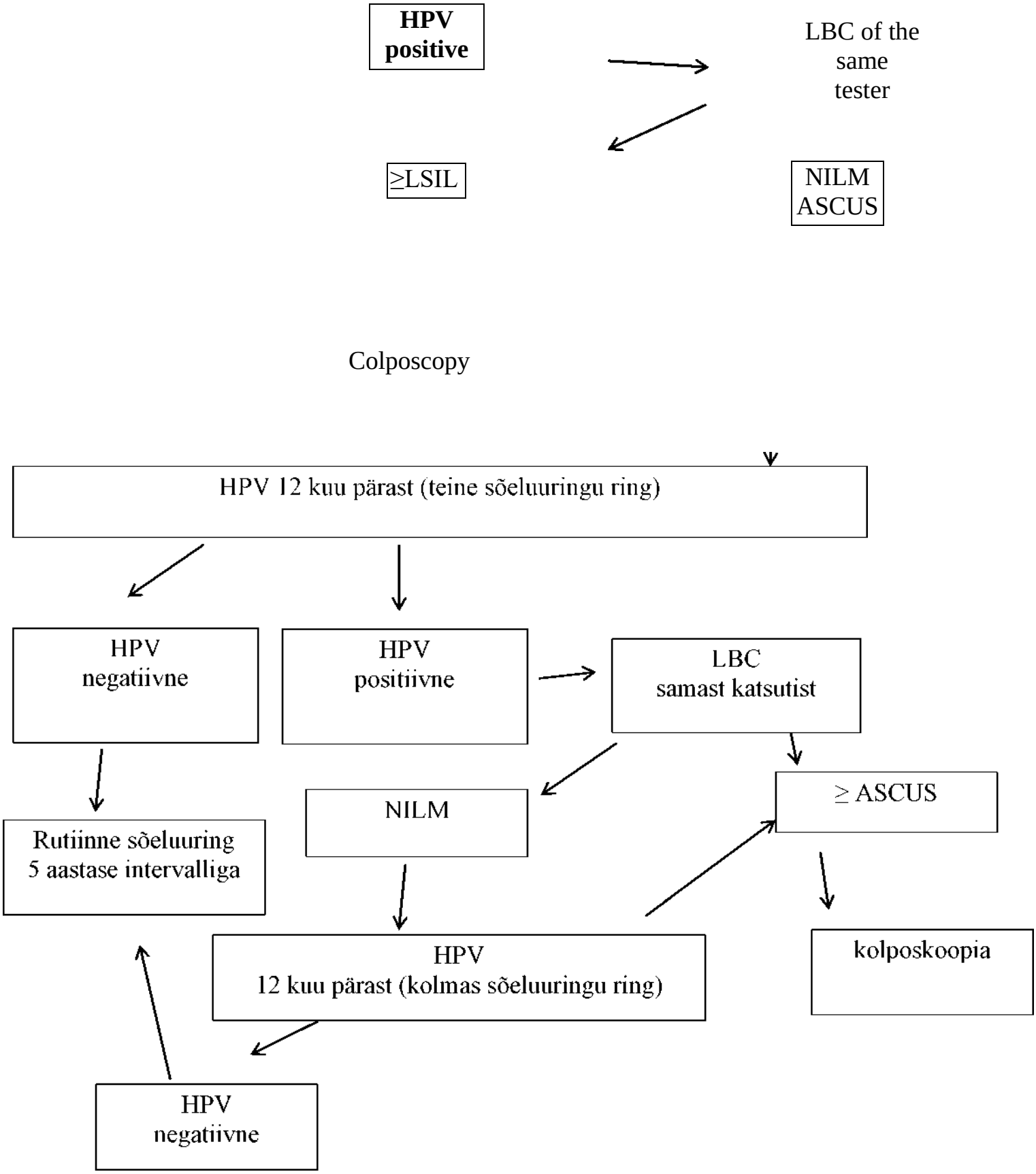


Figure 7: Monitoring of women ≥30 years of age, in gynaecocytological response to LSIL

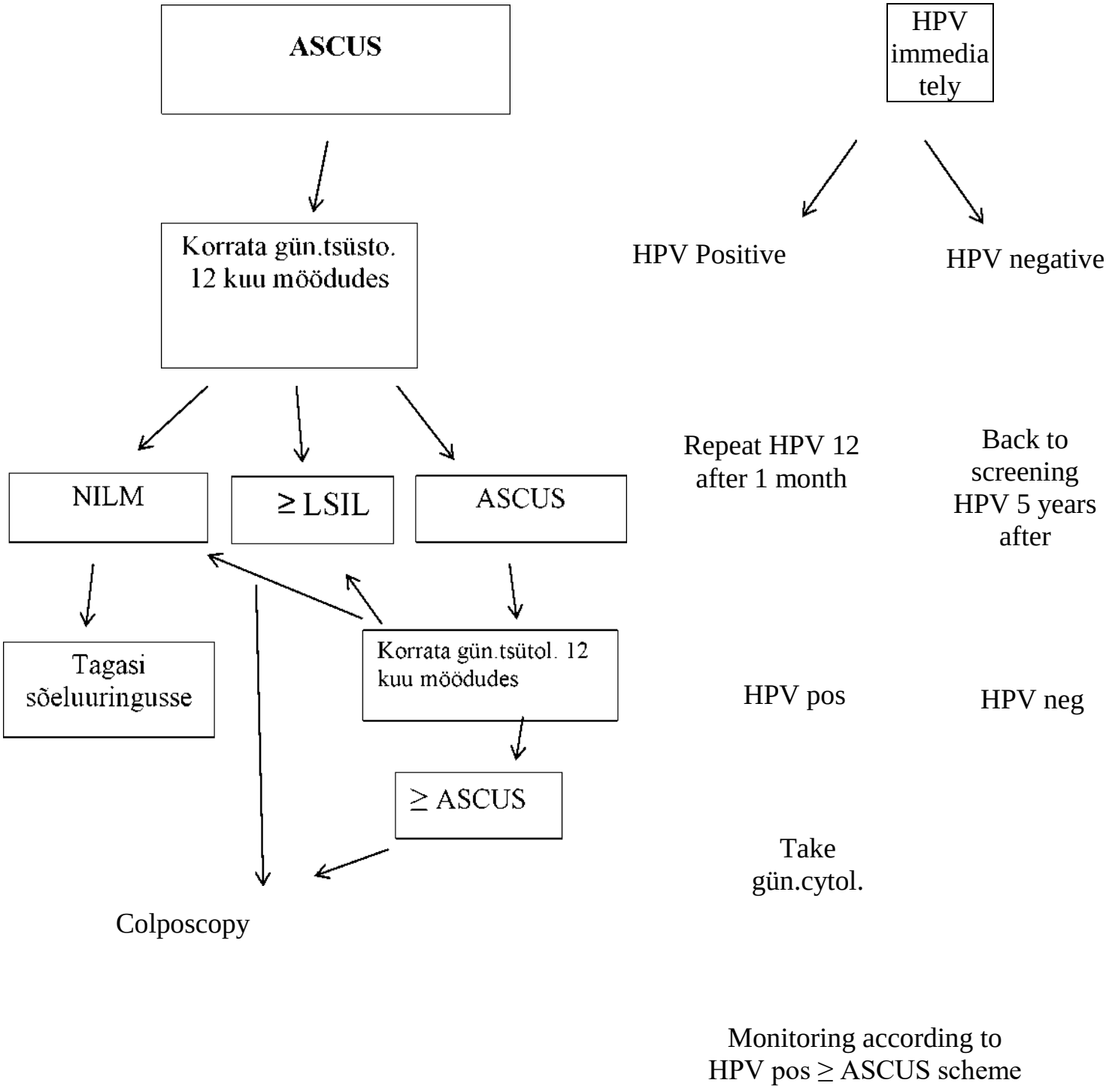


Figure 7: Monitoring of women ≥30 years of age, in gynaecocytological response to LSIL

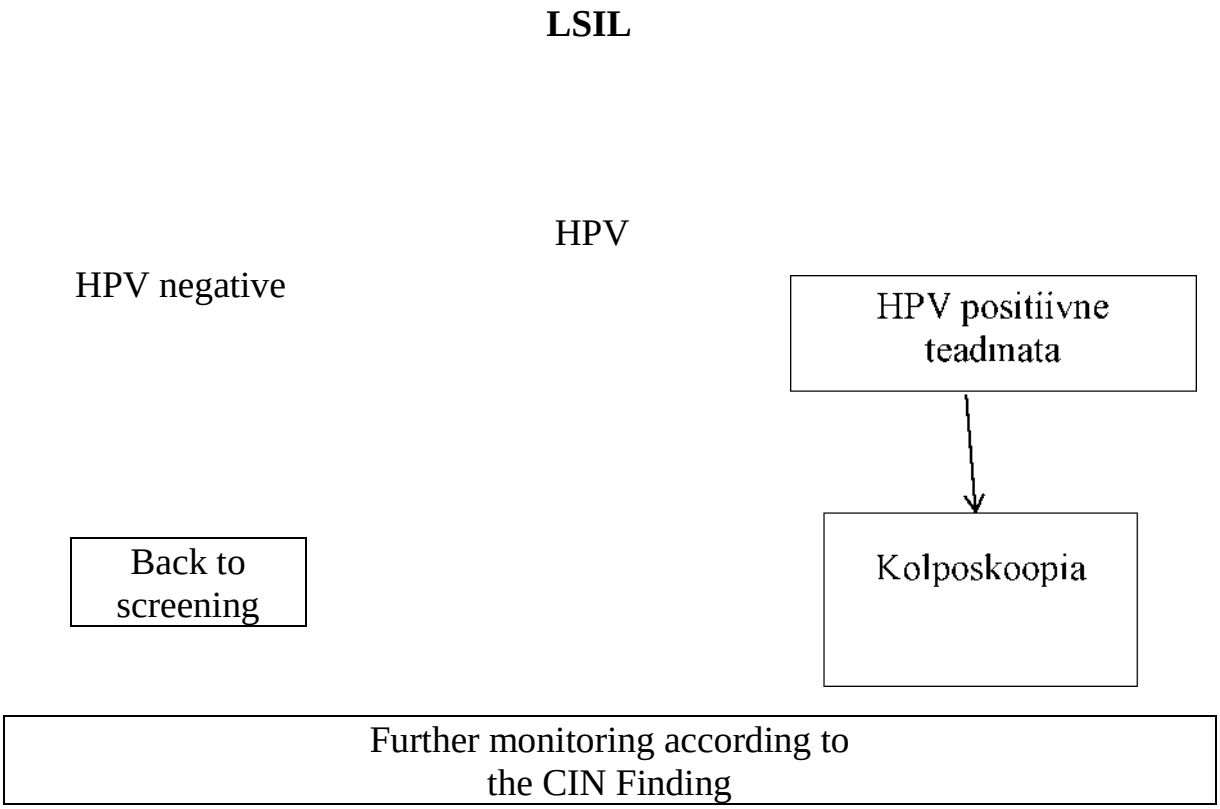


Figure 8: Monitoring of women 25-29 years of age, in response to gynaecocytological examination ASCUS or LSIL

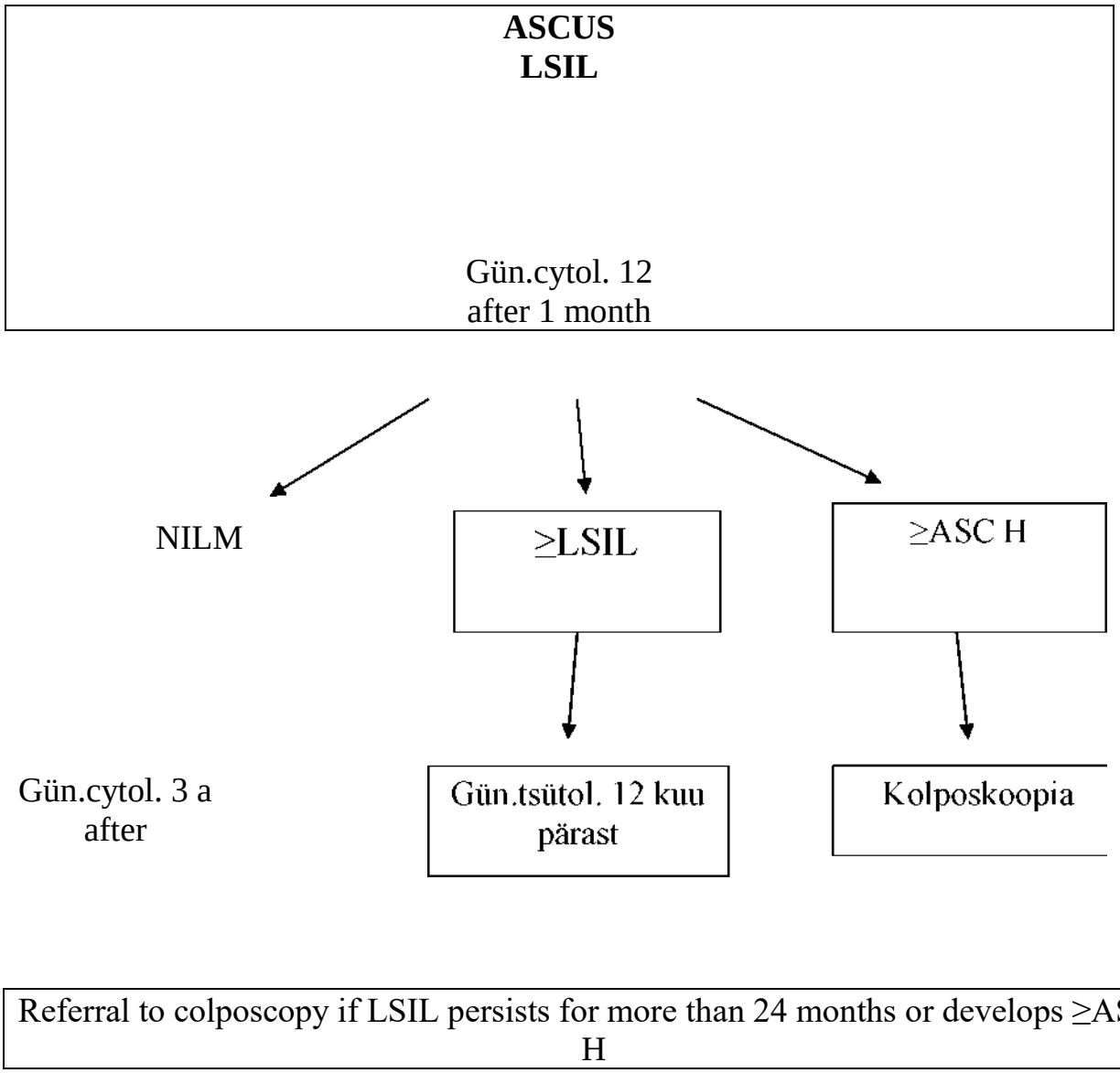


Figure 9: Monitoring of all age groups in gynaecocytological response for ASC-H and HSIL

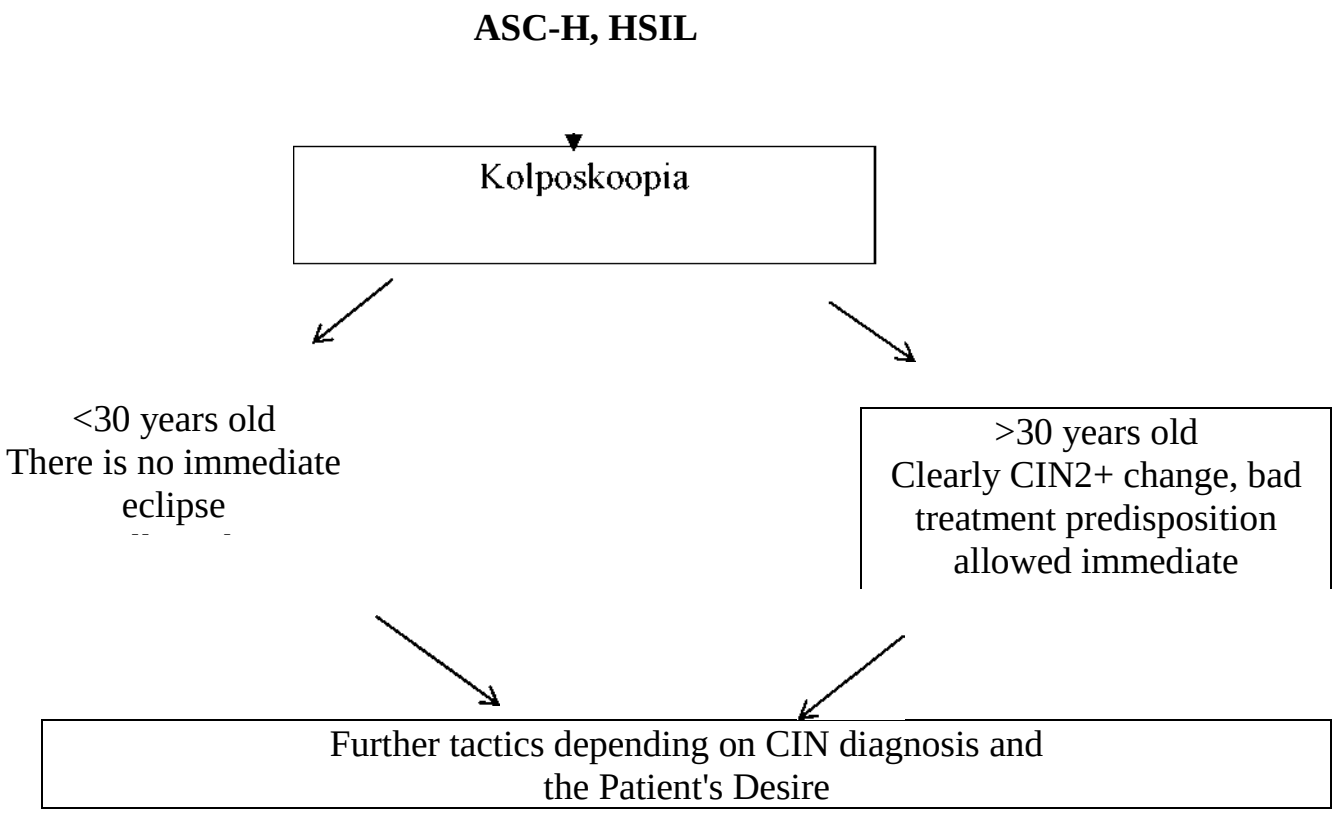


Figure 10: Observe all women for atypical glandular cells (all subgroups of AGC) in gynaecocytological examination response

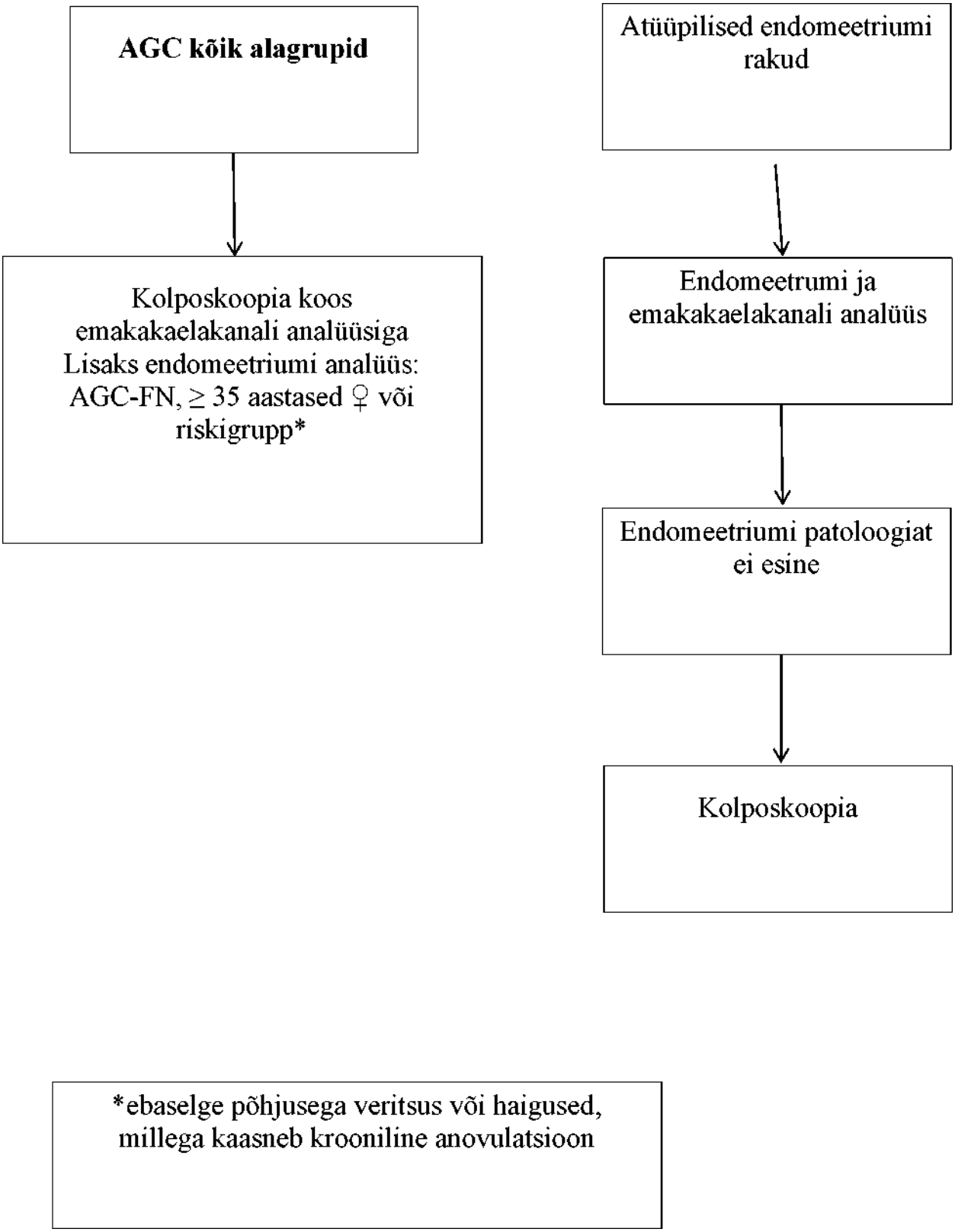


Figure 11: Further monitoring of the AGC

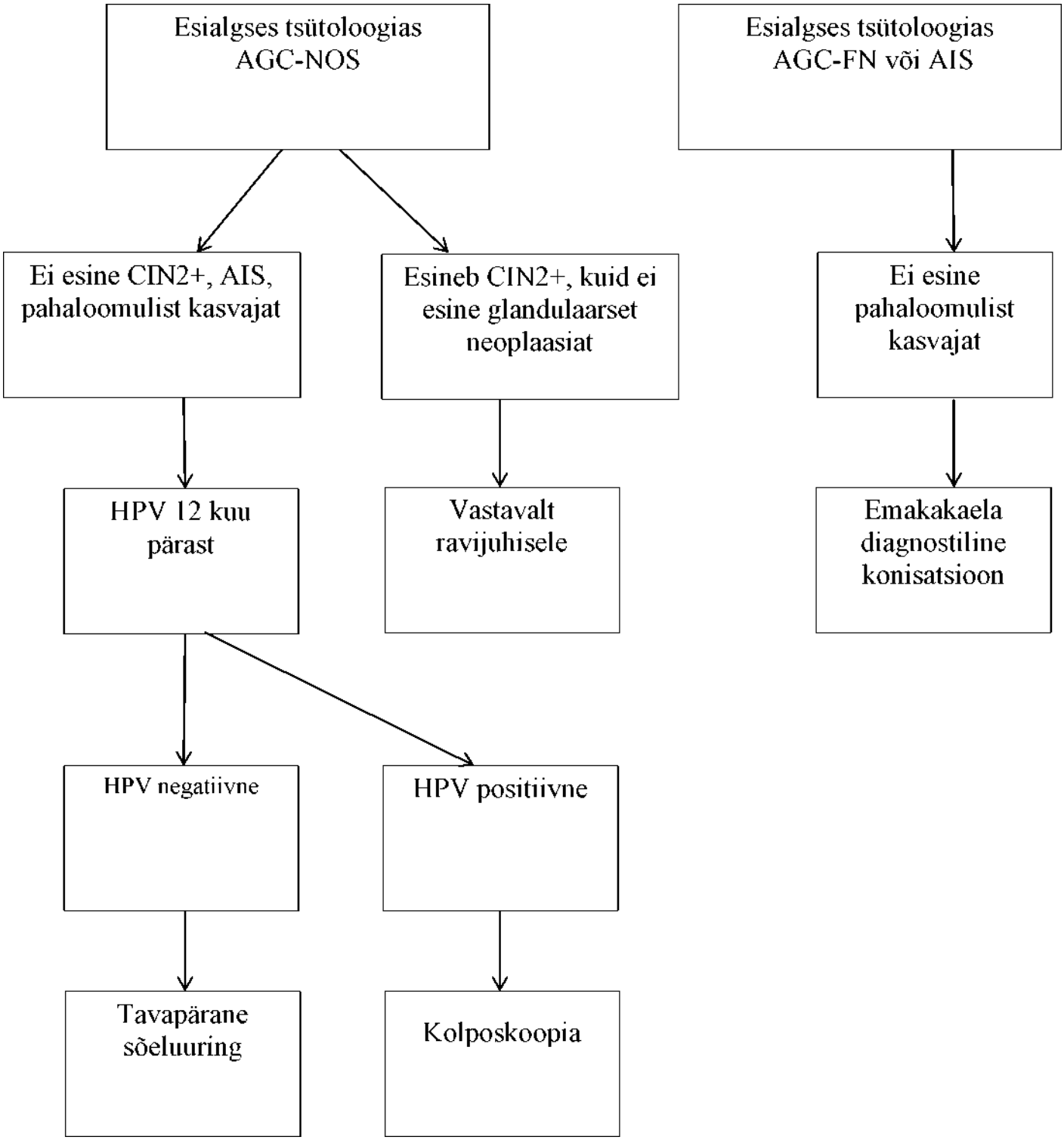


Figure 12: Monitoring of women with immunosuppressive conditions

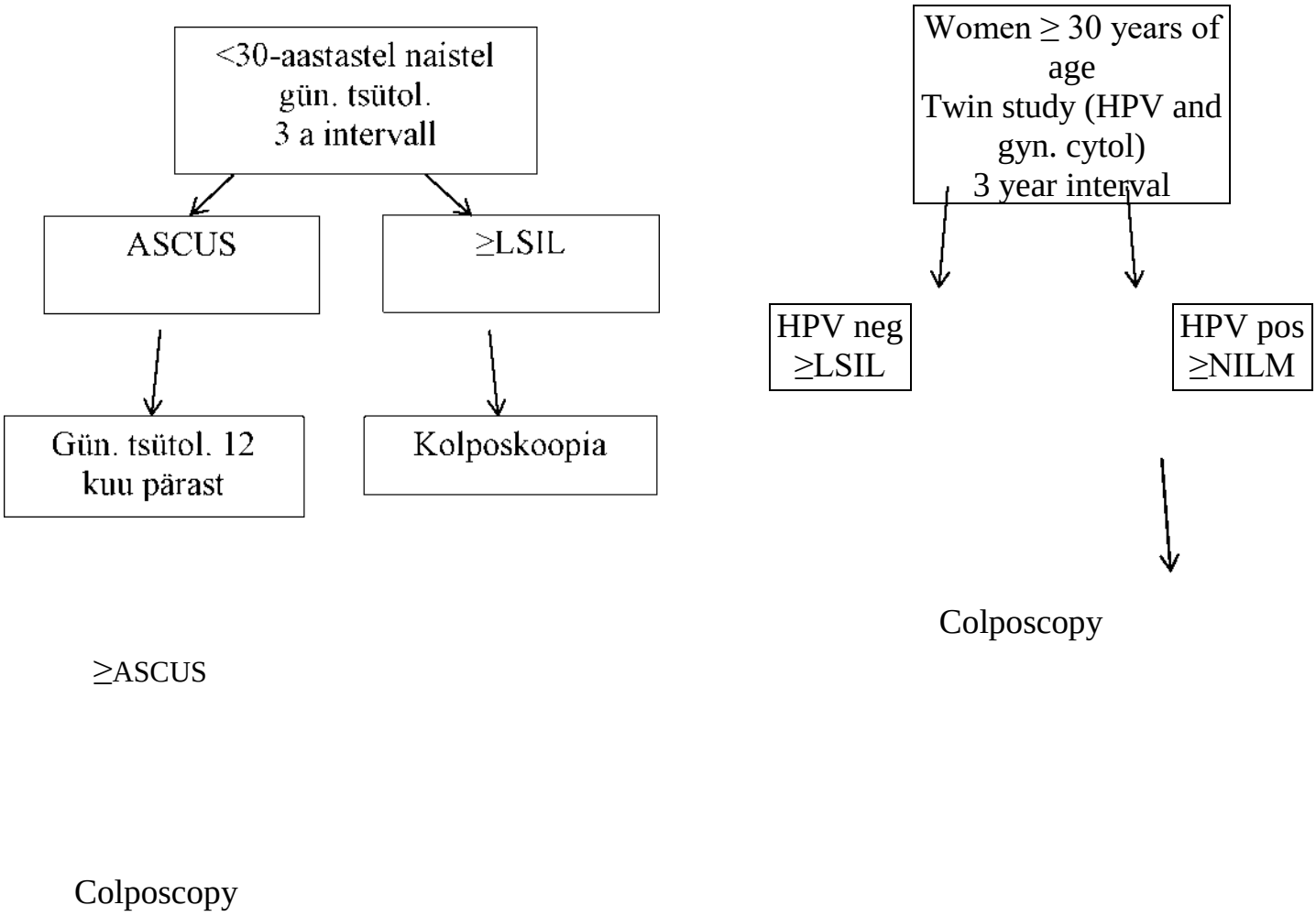
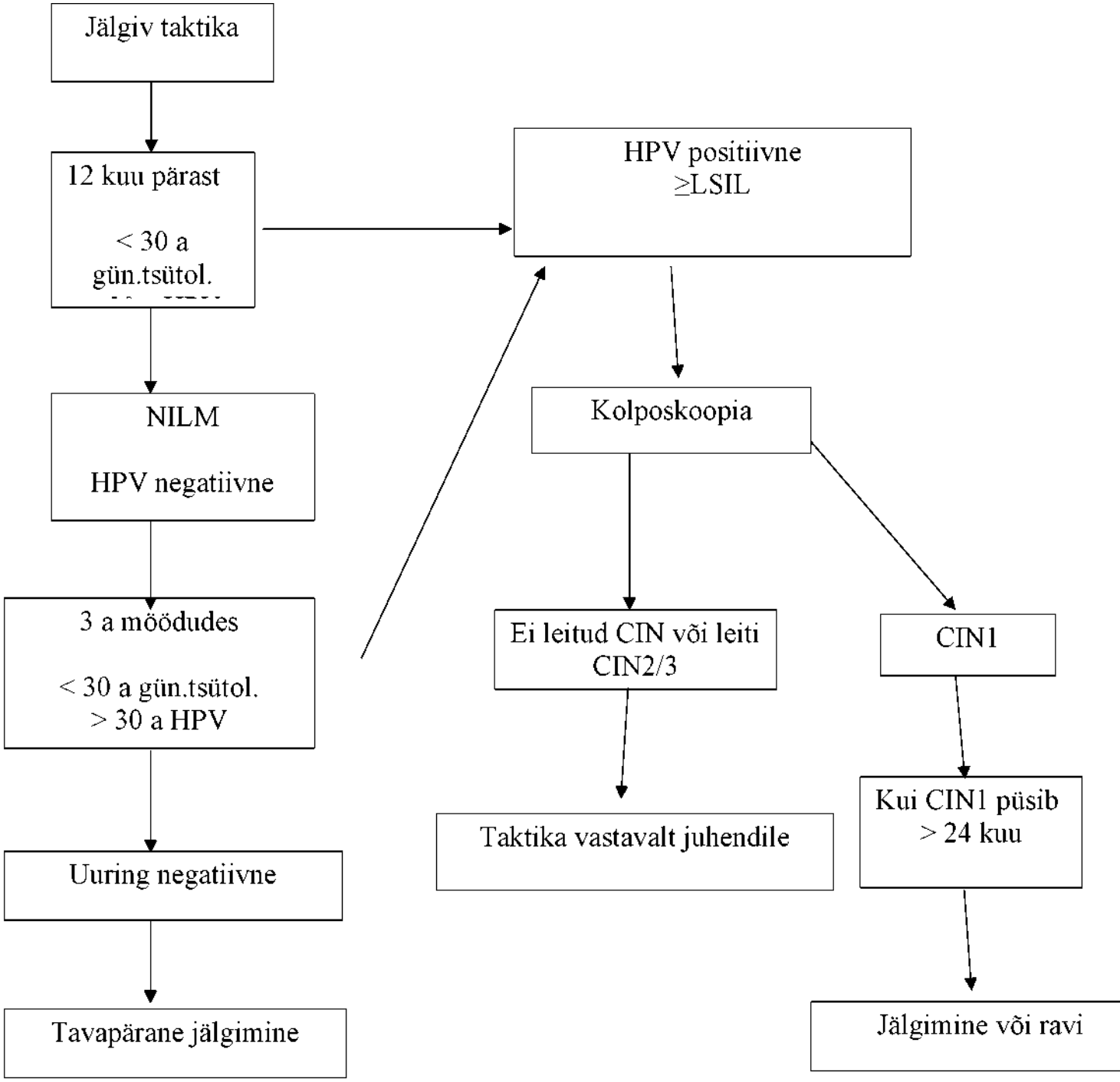
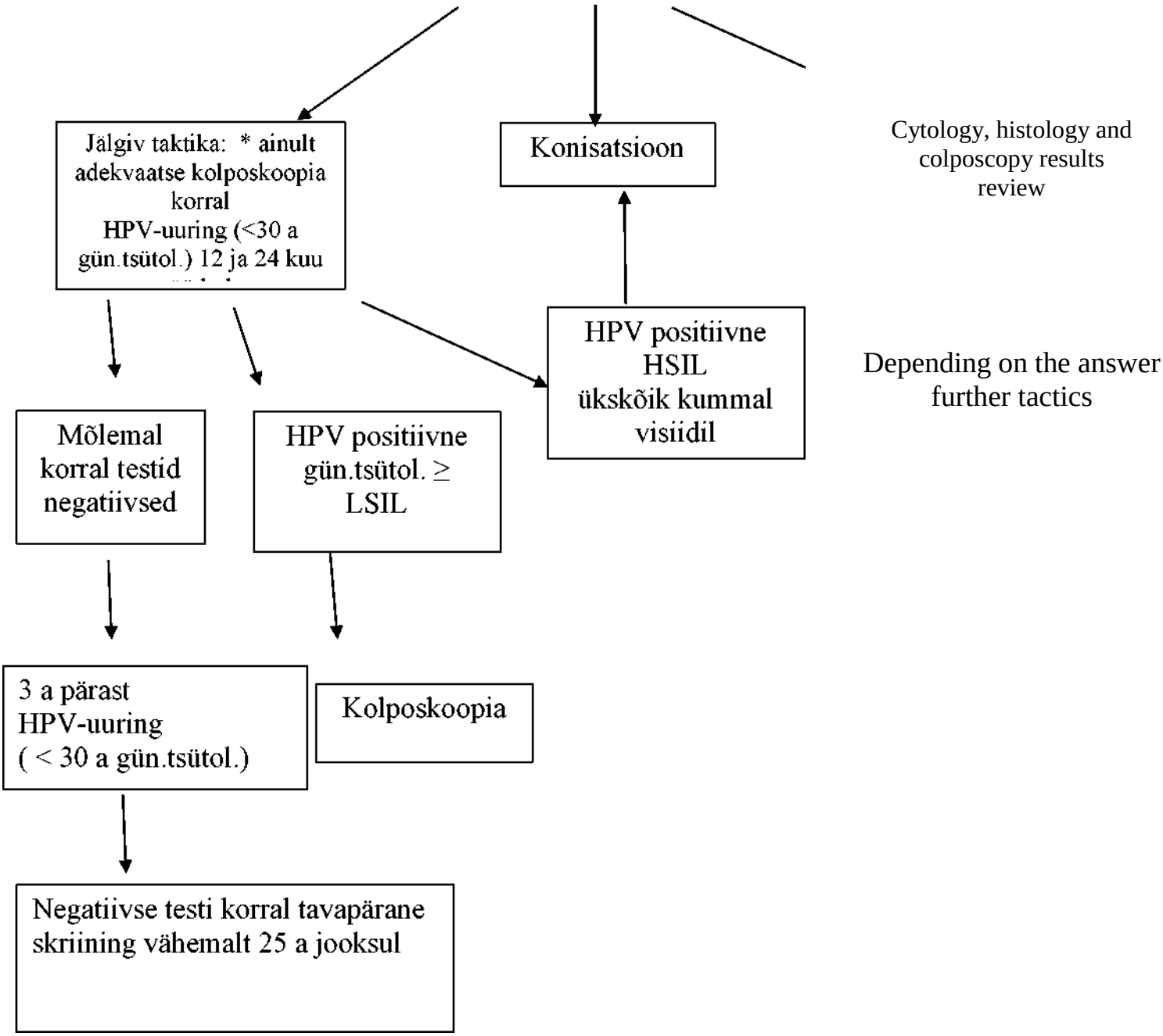


Figure 13: HPV Positive or Unknown, in Gynecocytology ASCUS, LSIL. Adequate colposcopy, histology ≤ CIN1



- Kui üle 30 a naisel osutub jälgimisperioodil HPV-uuring positiivseks, tuleb teostada ka günekotsütoloogiline uuring
- Kui gün.tsütol. uuringus leitakse jälgimisperioodil ASCUS, korrata vastavalt vanusele HPV-uuring või günekotsütoloogline uuring 12 kuu möödudes
- Jälgimistaktika valimise eelduseks on adekvaatne kolposkoopia. Kui TZ ei ole kolposkoopial nähtav, tuleb teostada ka emakakaelakanali analüüs

Figure 14: HPV positive or unknown, in gynecocytology ASC-H or HSIL. Adequate colposcopy, histology ≤ CIN1



If a woman over 30 years of age is tested positive for HPV during the follow-up period, a gynaecocytological examination should also be performed.

Adequate colposcopy is a prerequisite for choosing tracking tactics. If TZ is not visible on the colposcopy, cervical channel analysis should be performed. Diagnostic cervical retention is preferred in case of inadequate colposcopic findings

Figure 15: Monitoring of CIN2 in women who do not want cervical condensation due to potential risks to reproductive health

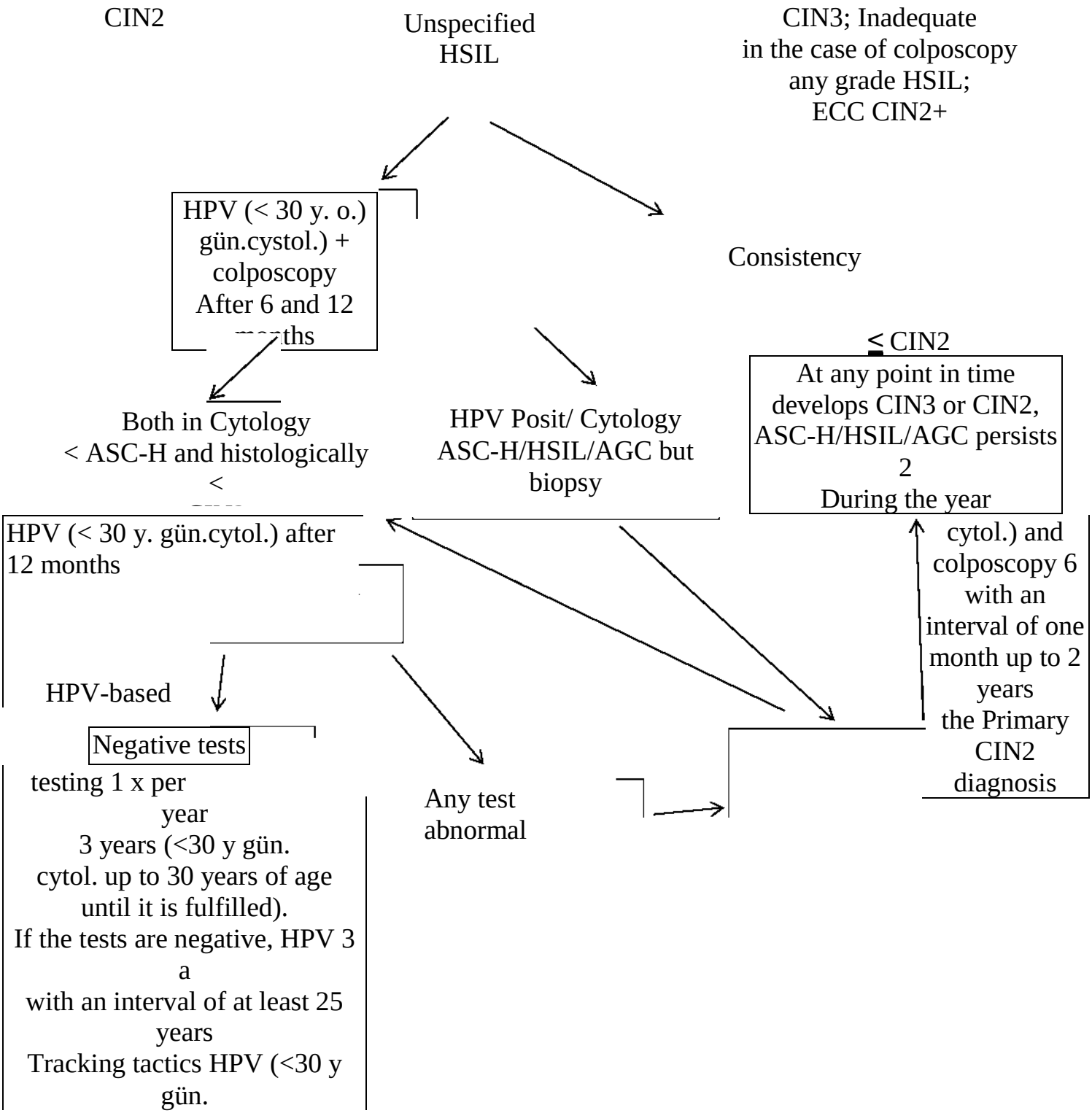


Figure 16: Vulva lesion treatment scheme

Suspicion of malignancy

There is no suspicion of malignancy

Biopsy, hearth photography, indicate place where the biopsy was taken

Direct immediately to oncogynecologist Do not make a mistake or

Clinically lichen sclerosus, lichen planus

Photo lesion later for comparison Treatment of high With hormone content glucocorticoid-

After 2 weeks, evaluate the effectiveness of treatment

In the absence of effect biopsy

_____ \lz ____

Clinically vulva HSIL, differentiated Vin

Biopsy, indicate biopsy taking location Without without histology do not of exile, prevents

Clinically vulva LSIL

where applicable later SLNB Advice Treatment as Symptoms or cosmetic for the purpose of:

2.2.Tables

2.2.1. Table 1: Reid's Colposcopy Index

http://www.ut.ee/ARNS/Guidelines/_ precancerous cervical conditions.pdf

Colposcopic identifier	0 points	1 point	2 points
Damage lesion boundary	Condylomatous or micropapillary surface drawing; inconclusive boundary; hairy or feathery boundary; jagged, angular, geographically contoured yellow; the performance of satellite lesions	Smooth, well-defined, regular symmetrical lesion.	Fog with raised or hatched edge; inside the hearth: the more pronounced atypical league is surrounded by a less pronounced atypical league and the boundary between them is clearly
Colour	Slightly coloured; reversible, blurred, translucent acetowhite; pure snow-white lesion with a bright surface.	Medium coloured; grey-white, glossy surface.	Matt, opaque white, oyster-white, gray.
Drawing of blood vessels	Fine puncture; fine mosaic; uniform, fine-diameter capillaries between micropapillary or microchonomylomatous drawings.	There are no blood vessels.	Coarse puncture; coarse mosaic; wide, irregular capillary gap; pathological expansion of individual capillaries.
0-2 points	Normal finding; immature metaplasia, changes in HPV		
3-4 points	CIN1; CIN 2		
5-6 points	CIN 2; CIN 3		

2.2.2. Table 2: Modified Reid Colposcopy Index

http://screening.iarc.fr/doc/colpochapterappendix.pdf

Colposcopic identifier	0 points	1 point	2 points
Damage lesion boundary	Condylomatous or micropapillary surface drawing; unclear boundary; A border of a furry or feathered nature; jagged, angular, geographically contoured gills; the performance of satellite lesions	Smooth, well-defined, regular symmetrical lesion.	Fog with raised or hatched edge; inside the hearth: the more pronounced atypical league is surrounded by a less pronounced atypical league and the boundary between them is clearly distinguishable.
Colour	Slightly coloured; reversible, blurred; translucent acetowhite; pure snow-white lesion with a bright surface.	Medium coloured; grey-white, glossy surface.	Matt, opaque white, oyster-white, gray.
Drawing of blood vessels	Fine puncture; fine mosaic; uniform, fine-diameter capillaries between micropapillary or microchonomylomatous drawings.	There are no blood vessels.	Coarse puncture; coarse mosaic; wide, irregular capillary gap; pathological expansion of individual capillaries.

Coloring with iodine	Positively coloured with iodine, giving mahogany brown color; negative coloration of insignificant lesions, e.g. yellow color on lesion, which has obtained ≤ 3 points according to previous criteria.	Partial dyeing with iodine; speckled appearance.	Negative colouring of significant lesions with iodine, e.g. lesions that have obtained ≥ 4 points according to previous criteria, will be coloured yellow.
0-2 points	CIN 1		
3-4 points	CIN1; CIN2		
5-8 points	CIN3		

2.2.4. Table 3: Swede colposcopy index

http://ifcpc.org/medical-professionals/swede-score/

Colposcopic identifier	0	1	2
Dyeing with acetic acid	None; weak, translucent	milky; cloudy	matte; highly coloured
Bulb outline and surface	none; diffuse	sharp; jagged; geographical satellites	sharp; above the surface; peeling
Vascular drawing	fine; regular	none	coarse; atypical
Bullet size	<5mm	5 to 15 mm or 2 quadrants	>15mm or 3-4 quadrants or endotservically unobservable

Coloring with iodine	brown	spotted or blurred yellow	Clearly yellow
0-4 points	mild grade change (LSIL)		
5-7 points	Severe Grade Change (HSIL), no suspicion of invasion		
8-10 points	Severe Grade Change (90% HSIL), Suspected Invasion		